

Pesticide fate under varying cropping systems and soil depths: A study using leaching experiments and inverse modelling.

Clémence Pirlot^{a,*}, Alodie Blondel^b, Boris Krings^b, Bastien Durenne^b, Olivier Pigeon^b, Aurore Degré^a

^a Uliège, Gembloux Agro-Bio Tech, TERRA Teaching and research centre, 2, Passage des Déportés, 5030 Gembloux, Belgium

^b CRA-W (Centre Wallon de Recherches Agronomiques), 11, rue du Bordia, 5030 Gembloux, Belgium

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ABSTRACT

Current pesticide leaching risk assessments overlook critical site-specific factors such as soil depth and agricultural practices. Relying on transport parameters from databases or manufacturer studies, often based on limited soil types, can lead to inaccurate contamination risk estimates and ineffective protection of groundwater resources. In this study, the fate of eight pesticides of concern for groundwater was investigated under three cropping systems and three soil depths. Leaching experiments were carried out in undisturbed columns from a loamy agricultural soil and mass balances were realized. Inverse dual-porosity modelling using Hydrus 1-D was then performed to adjust mobility parameters. The results reveal that different soil properties and structure between soil depths have a more substantial impact on pesticide leaching behaviour than the cropping systems. Significant differences in pesticide transport and retention are observed between soil horizons, illustrating the inadequacy of using surface parameters for the entire soil profile, which can lead to underestimation of groundwater contamination. Our analysis indicates that root architecture, soil properties and surface tillage can affect pesticide leaching dynamics. While short-term differences between cropping systems were limited, these factors could be important for long-term effects. The experimental transport parameters showed discrepancies with established databases, where higher adsorption and degradation could underestimate pesticide leaching and metabolites production. This study highlights the need to adapt transport parameter values for all pesticides of concern to site-specific conditions. In addition, accurate risk assessment requires advanced modelling techniques that take into account soil depth variability and local conditions to improve water protection decision-making. Future research should focus on long-term monitoring of the effects of sustainable agricultural practices on pesticide behaviour over several seasons and for a range of soil types. Special emphasis should be placed on the role of metabolites in environmental contamination.

1. Introduction

The current use of pesticides in agricultural systems has had a negative impact on the quality of natural resources. Pesticide use in agriculture and worldwide has increased by 50 % compared to the 1990s (FAO, 2022). Part of these pesticides will leach through the soil profile and lead to a deterioration of groundwater quality (Jurado et al., 2012; Labite et al., 2013). Despite a decrease in pesticide use in Europe, a large number of pesticides and their metabolites are frequently found in groundwater, which is the largest source of drinking water for many countries (Baran et al., 2021; EurEau, 2021; Kiefer et al., 2019; Köck-Schulmeyer et al., 2014; Pietrzak et al., 2019).

In Wallonia (southern Belgium), almost 80 % of the water supply, especially for drinking water, comes from groundwater. Between 2017 and 2020, pesticides were detected in 61.7 % of monitoring sites in measurable concentrations (SPW ARNE, 2022). Herbicides are responsible for the majority of pesticide contamination in groundwater. Among these, eight pesticides were still authorised when the research was carried out: bentazone, chlorotoluron, metamitron, metazachlor, S-metolachlor, terbuthylazine, flufenacet and ethofumesate. At the European level, the Water Framework Directive (2000/60/EC) and the Groundwater Directive (2006/118/EG) seek to limit groundwater contamination by requiring groundwater monitoring and by forcing Member States to define concentration limits for pollutants (European Commission,

* Corresponding author.

E-mail addresses: clemence.pirlot@uliege.be (C. Pirlot), a.blondel@cra.wallonie.be (A. Blondel).

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2000, 2006). However, inconsistencies between regulations are commonly found between soil and groundwater pesticides. As a result, worldwide regulations on soil pesticides do not take into account a sufficient number of pesticides with appropriate standard values to meet groundwater regulations and protect them effectively (Li, 2018). This is particularly the case in mixed rural and residential areas, where exposure to soil pesticides occurs indirectly. The soil-groundwater continuum needs to be properly considered in pesticide regulations (Li, 2021). To assess the risk of groundwater contamination by an active substance or its metabolites, leaching indices can be quickly calculated. However, many indices exist and the results obtained by different approaches can be contradictory (Akay Demir et al., 2019). Thus, solute leaching models relevant for the European Union are generally used to assess leaching potential (European Commission, 2014).

The two processes that predominantly control pesticide mobility within a soil profile are adsorption and degradation (Malla et al., 2021). Many factors will influence these processes such as pesticide properties but also soil characteristics, site conditions, and pesticide application practices (Arias-Estévez et al., 2008; Grodner et al., 2014). The solubility of a pesticide in water, which refers to its ability to dissolve in water, can also determine its leaching characteristics into groundwater and its potential to run off into surface water (Rasool et al., 2022). This parameter is well characterised conversely of the two parameters needed for modelling: the sorption coefficient (K_d) and half-life time (DT_{50}) (Bewick, 1994; Kahl et al., 2015; Katagi, 2013; Wauchope et al., 2002).

The FOCUS groundwater workgroup has developed nine scenarios based on a combination of weather, soil and cropping data to represent agriculture in the European Union (European Commission, 2014; FOCUS, 2009). In order to authorize or renew the authorization of a pesticide in the EU, the 9 FOCUS scenarios are used to model the leaching risk of the active compound for the whole geoclimatic range of Europe including sorption and degradation parameters derived from laboratory and lysimeter experiments. Most of these experiments are conducted by pesticide-producing companies, such as manufacturers or registrants, at several sites within a limited number of European countries. However, this approach can be misleading. For instance, Labite et al. (2013), showed significant differences between the pesticide leaching results given by the PELMO model parameterised for soil and weather conditions specifically for the site studied in Ireland and for the EU FOCUS scenarios used. The latter overestimated leaching by 42–99 %. Several studies show that pesticide properties such as K_d or DT_{50} provided by databases such as the PPDB are not always reliable for modelling pesticide fate in soils (Cueff et al., 2020; Mamy et al., 2024; Ullucci et al., 2022). They advise to determine these properties for the soil studied to have more accurate values. The FOCUS groundwater workgroup also recommends estimating the pesticide sorption and degradation parameters by inverse modelling of leaching experiments under site-specific conditions (FOCUS, 2009). However, in Belgium, pesticides transport parameters provided by manufacturers are currently still used to model the leaching risk of pesticides.

Nowadays, alternative and more sustainable agricultural practices are being developed, based on several principles such as reducing tillage, incorporating crop residues into the soil, maintaining soil cover and diversifying crops within the rotation (Verhulst et al., 2010). These practices can affect the temporal dynamics of soil hydraulic properties such as water retention and other factors such as soil structure or organic carbon content (Pirlot et al., 2024). Different crops in rotations will affect soil structure through their different root systems (Hu et al., 2009). For example, Lichter et al. (2008) found significantly larger macro-aggregates in the soil under wheat because wheat has a more horizontal root system than maize and the higher density of plants results in a denser shallow root system. Different crops also generate different soil microbial communities, with roots releasing a variety of different exudates depending on the crop. In general, earthworm abundance, diversity and activity increase under conservation agriculture (Chan, 2001; Fox et al., 2022; Lori et al., 2023). Therefore, pesticide

leaching within a soil profile is influenced by soil properties as well as land use, land management and crop characteristics (Alletto et al., 2015). As a result, the K_d and DT_{50} values found in databases are generally specific to the pesticide for a given soil type at a given depth and for a given set of agricultural practices (Imig et al., 2023a; Weber et al., 2004).

Cueff et al. (2020) showed that vertical transfer of water and pesticides in undisturbed soil columns was more pronounced in conservation agriculture production system than in conventional due to greater pore connectivity. In this context, the main risk is that preferential flows are occurring and generating a rapid vertical transfer of pesticides. Currently, the understanding of the effect of agricultural practices on pesticide leaching is unsatisfactory, which limits our ability to predict the environmental impact of cropping systems (Marín-Benito et al., 2018). Most studies try to isolate the effect of specific agricultural practices, such as tillage. However, it would be more appropriate to consider the effect of all practices to assess and compare pesticide transport under several different production systems (Willkommen et al., 2021).

The adsorption and degradation processes of pesticides will also vary with the depth of the soil. The different layers will have different textures, organic carbon contents, temperatures, pH, structure and fauna (Katagi, 2013). Available data on pesticide adsorption and degradation concern predominantly the surface soil layers and few studies are performed on the deeper soil layers. Pot et al. (2010), found different pesticide behaviors within the first three horizons of a sandy-loamy soil. Adsorption of chemicals to soil organic matter is the dominant adsorption mechanism when the organic matter fraction of the soil is higher than 2 % (Arienzo et al., 1994). Adsorption will also depend on the clay and metal oxide content (Weber et al., 2004). The organic carbon content of soils tends to decrease with depth, making adsorption generally lower in deeper soil layers (Katagi, 2013). In addition, the microbial population and temperature generally decrease with depth, making biodegradation slower (Mills et al., 2001). The composition of the soil microbial community will change with depth and the abundance of some bacteria such as Gram-positive will increase with depth (Fierer et al., 2003). Moreover, earthworms, in varying abundance in different soil layers, will impact the physical properties of the soil by increasing the water infiltration rate and the macroporosity (Johnson-Maynard et al., 2007).

To investigate the movement of pesticides within a soil profile, intact soil column leaching experiments are commonly used as they allow to get closer to field conditions than laboratory batch experiments (Katagi, 2013; USEPA, 2008). Observed data such as water and solute fluxes are then used to estimate pesticide transport parameters by inverse modelling (Dusek et al., 2015; Kahl et al., 2015; Kondo et al., 2020; Sniegowski et al., 2009; Sur et al., 2022). These parameters may be difficult to obtain by field leaching experiments due to soil heterogeneity and complex boundary conditions (Dusek et al., 2011). Preferential flows will have a dominant role in the transport of water and contaminants in structured soils (Jarvis, 2007; Köhne et al., 2009; Varvaris et al., 2021b). The preferential pathways, mainly macropores of biological origin such as earthworm burrows, root holes and of mechanical origin such as cracks, can rapidly transport pesticides deeper and increase the risk of groundwater contamination. Different approaches can be used to take into account the heterogeneity of the matrix and the preferential flows, such as the dual-porosity approach (DP) or the double permeability approach (DPM) (Šimůnek et al., 2003). Imig et al. (2023a) determined the transport parameters of four pesticides by inverse modelling using Hydrus 1-D with herbicide concentrations in drainage water. They showed that peak concentrations in the presence of preferential flows were correctly modelled using the DP model. Hydrus 1-D was also successfully used by Pot et al. (2005) and Köhne et al. (2006) to model water and pesticide fluxes within a soil column.

Therefore, the objectives of this study are

- (i) to assess the fate of eight pesticides, problematic for groundwater, using leaching experiments to obtain breakthrough curves in intact columns from a loamy agricultural soil;
- (ii) to highlight the sensitivity of cropping systems and soil depth on the mobility parameters obtained by inverse modelling;
- (iii) to compare those parameters with the theoretical parameters of the manufacturers, those currently used in Belgium and with the ranges of values given in the Pesticides Properties DataBase for Europe.

2. Materials and methods

2.1. Soils and cultivation practices

Soil columns were collected on an agricultural soil located on the EcoFoodSystem experimental plots (50°56'50.6"N, 4°70'93.2"E) of the Terra teaching and research centre of the University of Liege in Gembloux, Belgium. Within this experiment, three different agricultural production systems have been implemented in September 2020. These systems are specifically designed to challenge and differ from conventional production systems. They integrate multiple sustainable agricultural practices such as 8-year rotations, pesticide-free agriculture, and periods of sheep grazing. The rotations have been thought out to meet the food needs of the future, based on a study carried out by EAT LANCET (Eat Lancet Commission, 2019). The first system is called 'reference'. Animals are integrated into this system through the flow of agricultural by-products and manure. The second system is called 'Integrated Crop Livestock System (ICLS)'. Here, animals are integrated through grazing periods. The third rotation named 'vegan' is based on a future where animals are no longer part of the system. The rotation is designed to meet human nutritional needs. The 'vegan' designation is derived solely from the use of crops intended exclusively for human consumption and the absence of inputs of animal products such as manure. The three systems are implemented on adjacent 18 × 84 m plots.

The climate is temperate oceanic with an average annual temperature of 10.2 °C and an average annual rainfall of 837 mm. The soil is cutanic luvisol according to the World Reference Base for Soil Resources (FAO, 2015). The soil consists of a light brown Ap horizon that is ploughed up to about 30 cm. From 30 to 60 cm a lighter yellowish brown horizon AB is observed. After 60 cm there is an accumulation horizon of

light ochre clay Bt up to about 100 cm. The physico-chemical properties of the three horizons and the three cropping systems studied are given in Table 1. The notation is as follows: R = reference, I=ICLS, V = vegan, RH1–3 = horizon 1 to 3.

In July 2020, the winter wheat was harvested and stubble ploughed. Cover crops were then planted in September 2020 until the first crops of the experiment. In the first year of the rotations in 2021, sugar beet, maize, and camelina were planted in the reference, ICLS and vegan systems respectively. Tillage and soil preparation were performed on all rotations in April 2021. In the reference system, a manure application and sowing of beets at 100,000 plants/ha was done at the end of April 2021. Then, a hoe passage was carried out at the beginning and middle of June 2021 to remove weeds mechanically between crop rows, loosening the soil surface, while avoiding the use of chemical herbicides. In the ICLS system, the application of manure and the sowing of grain corn at 100,000 plants/ha were carried out in mid-May 2021. The rotary hoe was conducted in mid-June and the hoe in late June and early July 2021. In the vegan system, camelina sowing at 3,000,000 plants/ha was completed in late April 2021. The rotary hoe was run in mid-June 2021. Harvesting and stubble plowing of the camelina was completed in early September 2021. Fig. 1 shows the agricultural itineraries of the different experimental plots.

As the cropping systems are pesticide-free, there have been no pesticide treatments since September 2020. The most recent pesticide uses studied were S-metolachlor in 2018, ethofumesate and metamitron in 2016 and metazachlor in 2013.

2.2. Undisturbed columns

A total of 15 intact soil core columns were taken using 40 cm long Plexiglas tubes with an internal diameter of 8.4 cm. Porosity was calculated from the bulk density for each column and the density of the soil particles and organic matter, assuming a soil particle density of 2.65 g.cm⁻³ and an organic matter particle density of 1.4 g.cm⁻³. These values have been used because they correspond to common values for silt and soil organic matter (Porfiri et al., 2015; Ruehlmann, 2020; Schjønning et al., 2017).

All columns were collected between September 8 and 13, 2021. Nine columns were collected in the reference innovative system where sugar beets were implanted. The columns were taken from the first three horizons, i.e. three from 0 to 30 cm, three from 30 to 60 cm and three from 60 to 90 cm in the reference system. The three horizons were compared only for the reference system, as agricultural practices mainly influenced the surface horizon. Then, three columns were taken between 0 and 30 cm in the ICLS system where maize was established and three others between 0 and 30 cm in the vegan system where camelina had just been harvested. The columns were all driven in by hand. A wooden plate was placed over the columns with a piece of textile. Using a sledgehammer, they were gradually pushed into the ground to the desired depth. The surrounding soil was then removed with a shovel. Once the column was uncovered, the soil underneath was cut off with a trowel to extract it. Immediately after excavation, the ends of the columns were sealed. The columns were secured in a box to bring them to the laboratory on foot to minimize disturbance to the soil structure (Singh et al., 2002).

At 2 cm from the bottom of the tubes, a perforated Plexiglas plate was fixed to elevate the columns and allow water to pass through. A nylon textile disc was placed between the soil and the perforated plate to prevent soil particles from being washed away. A tube was placed in the water collection area and was connected to two Erlenmeyer flasks and a vacuum pump to provide a suction of 35 kPa to get closer to the field conditions (Dusek et al., 2015). All columns were built in triplicate.

To limit preferential flow along the column wall, paraffin was poured over the interface between the soil and the Plexiglas wall. The paraffin ensures that the water infiltrates the soil and does not flow along the wall (Arthur et al., 1998; Isensee and Sadeghi, 1992). The columns were kept free of vegetation for the duration of the experiment. The columns

Table 1

Physico-chemical properties of the soils for the three cropping systems (RH1, IH1 and VH1) and the three soil depths (RH1, RH2 and RH3).

	RH1	IH1	VH1	RH2	RH3
Depth	0–30	0–30	0–30	30–60	60–90
Crops	Sugarbeets	Maize	Camelina	Sugarbeets	Sugarbeets
Texture	Fine silty clay				
Clay (%)	13,8	12,3	17,2	13,1	19,2
Silt (%)	78,3	79,7	76,5	80,4	74,5
Sand (%)	7,9	7,9	6,3	60,4	6,2
Organic carbon (g/kg)	12,3	11,5	11,7	5,24	3,35
Cation exchange capacity (CEC) (meq/100 g)	9,4	8,8	10,6	8,3	10,6
pH water	7,6	7,6	7,6	7,8	7,9
Bulk density (g/cm ³)	1,41 ± 0,03	1,64 ± 0,01	1,59 ± 0,04	1,54 ± 0,01	1,64 ± 0,01
Porosity	0,462 ± 0,009	± 0,375	0,395 ± 0,015	0,415 ± 0,005	0,378 ± 0,002

	2020										2021														
	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Janv	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				
RH1	Winter			⚡		Cover crops									⚡				Sugar beets						⚡
IH1	Wheat			⚡	Maize										⚡										
VH1				⚡	Camelina														⚡						

Fig. 1. Technical agricultural itineraries for the three cropping systems studied.

were gradually saturated with distilled water to ensure uniform wetting (Kördel et al., 2008; Van Beinun et al., 2006). They were then drained freely without vacuum pump for 24 h to achieve field capacity and minimize differences in water content between columns at the beginning of the experiment (Weber et al., 2003). Because pesticides are photo-sensitive, the complete experimental set-up (columns, connections, and collecting Erlenmeyer flasks) was covered with aluminium (Hua et al., 2009).

The columns were gradually saturated with distilled water to ensure uniform wetting and avoid preferential flow pathways that could influence pesticide leaching dynamics (Kördel et al., 2008; Van Beinun et al., 2006). They were then drained freely without a vacuum pump for 24 h to achieve field capacity, simulating realistic soil moisture conditions and minimizing differences in water content between columns at the beginning of the experiment (Weber et al., 2003).

2.3. Pesticides properties

Eight herbicides with contrasting properties were studied. These eight herbicides were chosen because they are very commonly used in agriculture. Collectively, they account for 20 % of herbicide use in Wallonia (CORDER, 2020). In addition, they are all of different types (except metazachlor and S-metolachlor) with a wide range of solubility (6.6 to 7112 mg/l) and adsorption and degradation properties. Pesticides were used in formulation as they are applied in the field. Commercially formulated pesticides will have different mobilities and may be leached from soil columns to a greater extent than active molecules (Hua et al., 2009; Khan and Brown, 2016). The main properties of the eight herbicides from the Pesticides Properties DataBase (PPDB) (Lewis et al., 2016), as well as the recommended dose and concentrations applied to the columns are summarised in Tables 2 and 3.

Table 2

Main properties of the eight pesticides studied, based on the Pesticide Properties DataBase (PPDB).

Herbicide	Type	Crops	Action	Solubility in water at 20 °C (mg/l)	Octanol-water partition coefficient (pH 7, 20 °C)	pKa at 25 °C	Henry's law constant at 25 °C (Pa. m ³ /mol)	Persistency	Mobility	GUS index
Bentazone	Thiadiazine herbicide	Corn, rice, alfalfa, linseed, beans, ...	Inhibits photosynthesis (photosystem II)	7112	−0,46 (low)	3.51 (Weak acid)	7.20*10 ^{−5} (Non-volatile)	non-persistent	Mobile	1,95
Chlorotoluron	Urea herbicide	Cereals, potatoes, maize, vegetables	Inhibition of photosynthetic electron transport	76	2.5 (Low)	No dissociation	9.07*10 ^{−7} (Non-volatile)	Moderately persistent	Moderately mobile	2,01
Metamitron	Triazinone herbicide	Sugar beet, Fodder beet, Red beet	Inhibits photosynthesis (photosystem II)	1770	0.85 (Low)	No dissociation	8.95*10 ^{−8} (Non-volatile)	Moderately persistent	Moderately mobile	2,16
Metazachlor	Chloroacetamide herbicide	Oilseed rape, maize, sugarcane, cotton, ...	inhibition of cell division	450	2.49 (Low)	No dissociation	5,9*10 ^{−5} (Non-volatile)	Non-persistent	Mobile	1,75
S-Metolachlor	Chloroacetamide herbicide	Corn, soybeans, sorghum, potatoes, ...	inhibition of cell division	530	3.4 (High)	No dissociation	2.40*10 ^{−3} (Non-volatile)	Moderately persistent	Moderately mobile	2,32
Terbuthylazine	Triazine herbicide	Maize, Sorghum, Apples, Citrus, ...	Inhibits photosynthesis (photosystem II)	6.6	3.4 (High)	1.9 (Weak basic)	2.30*10 ^{−3} (Non-volatile)	Moderately persistent	Moderately mobile	2,19
Flufenacet	Anilide herbicide	Corn, Soybeans, Winter wheat, Sunflowers, ...	inhibition of cell division	51	3.5 (High)	No dissociation	1.30*10 ^{−3} (Non-volatile)	Non-persistent	Moderately mobile	2,49
Ethofumesate	Benzofuran herbicide	Sugar beet, Fodder beet, Red beet, onions, carrots, ...	Inhibition of lipid synthesis	50	2.7 (Moderate)	No dissociation	3.72*10 ^{−3} (Non-volatile)	Non-persistent	Moderately mobile	3,04

Table 3

Adsorption and degradation properties of the eight herbicides from the PPDB and concentrations used for the 5 ml pulse applied to the top of the columns. K_f is the Freundlich adsorption coefficient, n is the exponent from the Freundlich equation and DT_{50} is the half-life of the pesticide.

Herbicide	Formulation	K_f (l/kg)	1/n (β)	DT_{50} in field (days)	Recommended field application (g/ha)	Concentration of the solution applied on soil columns (mg/l)
Bentazone	Basagran SG	0.97	0.93	7.5	960 (peas)	92,7-109,6
Chlorotoluron	Chlorotoluron 500 SC	1.3	0.877	12.5	1500 (wheat)	141,1-156,2
Metamitron	Goltix 700 SC	1.09	0.81	11.1	3500 (sugar beet)	354,4-373,0
Metazachlor	Butisan S	1.02	0.993	6.8	750 (rapeseed)	57,6-84,7
S-Metolachlor	Dual Gold	0.93	0.888	21	1248 (sugar beet)	148,3-149,0
Terbuthylazine	Andes	5.1	0.93	21.8	750 (maize)	85,7-95,5
Flufenacet	Andes	4.38	0.92	39	450 (maize)	40,8-46,1
Ethofumesate	Ethomat 500	1.74	0.91	37.8	1000 (sugar beet)	65,8-125,9

2.4. Leaching experiment

An initial solution of 1 l of pesticides was prepared. The concentration of pesticides in the solution was calculated to meet the maximum recommended doses per hectare (Table 3). Since bentazone is in granular form, it is added first to the 1 l flask already two-thirds filled with deionized water. The other pesticides, in the form of a suspension concentrate, are added next. This order is recommended to limit the incompatibility of pesticides during mixing and to optimise the dissolution (BASF, 2017). A sample of this initial 1 l pesticide mixture is analysed to quantify the exact pesticide concentrations. The solution is stored at 4 °C and protected from light.

A 5 ml (0,9 mm) pesticide pulse is then applied to the column using a graduated pipette. The pulse is added drop by drop manually avoiding the outer circle of the column to avoid preferential flows. Then, 5 ml of rinse is added in the same way (Köhne et al., 2006; Van Beinum et al., 2006). The 5 ml of pulse and rinse together represent 1.8 mm of rain, which corresponds to a continuous light rain for 1 h. The top of the column is sealed with plastic film to limit evaporation. The column was then left to stand for 24 h. The leaching experiment was conducted under unsaturated conditions to simulate the situation in the vadose zone and under transient flow to imitate rain events. Water was added above the column at regular intervals. The first three additions of water were of 200 ml (36 mm) each. The next additions were 400 ml (72 mm) unless there was still water at the top of the column. The chosen water inputs to the columns represent possible intense rainfall for the area under study. During intense rainfall, between 30 and 80 mm can be expected to drop (IRM, 2020). The two volumes applied to the columns correspond to a heavy rainfall of 2 mm/min lasting 20 and 40 min respectively. In this case, the amount of water added depended on the amount of water still present. Eluted water samples were taken before each water application to the columns. The last samples were combined to create composite samples and to obtain the end of the breakthrough curve.

The duration of the experiment from the pulse to the last collected sample was the same for all columns, i.e. 57 days. This duration was chosen to obtain breakthrough curves in their descending part for all the pesticides and to model them more accurately. At the end of the 57-day experiment, the columns were directly disassembled and the soil was homogenized. Soil samples from each column were taken for the analysis of the residual pesticide concentrations. The percentages of pesticides eluted in the leachates (recovered) and adsorbed within the columns, in relation to the quantity applied above the column, were calculated. The mass balance is then calculated to determine the quantity of pesticide degraded by subtracting the quantity recovered and adsorbed from the amount applied to the column. However, it should be noted that with this methodology, the non-extractable fraction of pesticide residues can be included in the degraded fraction of the mass balance.

2.5. Pesticide concentration measurement

The initial mixture of pesticides was analysed using UPLC-DAD (Waters Acquity H class, USA). The column was an Acquity BEH C18 (50 mm × 2.1 mm (id) × 1.7 µm) with a flow rate of 0.8 ml/min. 1 µl of sample was injected. Mobile phase A was H₂O with 0.1 % formic acid and mobile phase B was acetonitrile. The gradient goes from 90 % aqueous phase to 100 % organic phase in 8 min, stays in 90 % organic phase for 1 min and goes back to 90 % aqueous phase for 1 min for system conditioning between injections.

After sampling at the bottom of the columns, water samples were stored at 4 °C in amber glass bottles. These samples were analysed within 7 days of collection. 2 ml of acetonitrile was added to 10 ml of sample, shaken manually and then centrifuged at 4800 rcf (relative centrifugal force) for 10 min at 4 °C before filtration through a 0.2 µm PTFE (polytetrafluoroethylene) filter. 10 µl was injected.

Water samples were analysed by liquid chromatography (Nexera X2™ Shimadzu, USA) coupled to a Time-of-Flight Mass Spectrometer (Q-TOF) (X500R ABSciex, Singapore). The column used was a Waters Acquity UPLC™ HSS T3 (100 mm × 2.1 mm i.d., 1.8 µm particle size) maintained at 40 °C with a mobile phase gradient (0.3 ml/min) consisting of: MilliQ water/methanol (90/10) and 2 mM ammonium formate acidified with 0.1 % formic acid in lane A and methanol acidified with 0.1 % formic acid in lane B. The gradient goes from 95 % aqueous phase to 100 % organic phase in 11 min, stays in 100 % organic phase for 4 min and goes back to 95 % aqueous phase in 0.5 min, then stays for 7 min for system conditioning between injections.

Soil samples were thawed and sieved 2 mm to homogenise and remove plant fragments, stones, etc. A 5 g sub-sample was then extracted using the QuEChERS method as follows: 5 ml of water was added, the sample was shaken and allowed to macerate for 30 min, then 10 ml of acidified acetonitrile (2 % formic acid) was added, the sample was shaken again and allowed to macerate for 30 min. A bag of QuEChERS salts (4 g MgSO₄, 1 g NaCl, 0.5 g sodium hydrogen citrate sesquihydrate, 1 g sodium citrate dihydrate) (Agilent, USA) was added and the sample was vortexed for 1 min at 20 Hz (Retch, MM400). After centrifugation at 4800 rcf for 15 min at 4 °C, the supernatant was filtered through a 0.2 µm PTFE filter and transferred to a glass vial. 5 µl was injected. Soil samples were analysed using the same analytical method as water samples, except for the elution gradient. It goes from 95 % aqueous phase to 100 % organic phase in 8 min, stays in 100 % organic phase for 3 min, goes back to 95 % aqueous phase in 0.5 min and stays in this composition for 7 min for system conditioning between injections.

The acquisition of results was performed in TOFMS mode in electrospray positive ionisation mode (ESI+) except for bentazone which was acquired in high resolution multi reaction monitoring mode (MRM_{hr}) in negative ionisation mode (ESI-). For quantification in water solution samples, a calibration curve was established using MilliQ water +20 % acetonitrile after ensuring that there was no matrix effect. Soil quantification was performed using a matrix-matched calibration curve with a reference material, i.e. biological soil without pesticide residues.

For each series of analyses, 3 spiked soil reference materials were extracted according to the protocol described above to verify the extraction efficiency. The spiked concentrations were 10, 1 and 0.2 µg/kg. Stability was checked for each injection sequence and could not exceed 20 % for each active ingredient.

The limit of quantification in water and soil is given in Table 4.

2.6. Water flow and pesticides transport modelling

Hydrus 1-D is a one-dimensional finite element model that simulates water, heat and multiple solute fluxes in unsaturated conditions (Šimůnek et al., 2008). All breakthrough curves were modelled using Hydrus 1-D. The hydraulic and transport parameters of the eight pesticides were obtained by inverse modelling. The units of length (L) are cm, the units of mass (M) are grams and the units of time (T) are days.

2.6.1. Inverse modelling

Inverse modelling is used to determine the parameters of interest from data obtained through soil column leaching experiments. Inverse modelling is based on numerical solutions of the Richards equations and the convection-dispersion equation using the non-linear Levenberg-Marquardt (LM) optimisation method (Marquardt, 1963). The LM algorithm minimises the differences between the observed and simulated values by iteratively adjusting the parameters of interest. This algorithm will optimise the parameters by minimizing the objective function $\phi(q, b)$ defined by (Šimůnek et al., 1998):

$$\phi(q, b) = \sum_{j=1}^m v_j \sum_{i=1}^n w_{ij} \left[q_j^*(x, t_i) - q_j(x, t_i, b) \right]^2 \quad (1)$$

where m is the number of different sets of measurements, n is the number of observations for measurement set j , $q_j^*(x, t_i)$ are the specific measurements at time t_i , position x , $q_j(x, t_i, b)$ are the corresponding model predictions obtained with the vector of optimised parameters b and v_j and w_{ij} are the weights associated with a measurement or point set.

Three statistical indicators were used to assess the performance of the inverse modelling and parameter optimisation: sum of squares (SSQ), root-mean-square error (RMSE) and coefficient of determination (R^2). Several studies suggest that both water flux data and solute concentration data are needed by models to provide a proper representation of solute transport (Coppola et al., 2009; Kahl et al., 2015; Kondo et al., 2020). A sequential or combined approach can be used, where the hydraulic and transport parameters are estimated successively or simultaneously from the observed water flux and concentration data in the eluates (Imig et al., 2023b; Imig et al., 2023a; Šimůnek et al., 2002). The sequential approach is commonly used because it prevents the uncertainty in the hydraulic parameters from propagating into the uncertainty of the solute transport parameters, resulting in a better fit (Abbasi et al., 2003; Kahl et al., 2015).

2.6.2. Water flow modelling

The flow of transient water within the columns was simulated by two

Table 4

Limits of quantification (LOQ) of the eight pesticides studied in water in µg/l and in soil in µg/kg.

Pesticides	Water LOQ (µg/l)	Soil LOQ (µg/kg)
Bentazone	0.01	0.15
Chlortoluron	0.02	0.2
Flufenacet	0.01	0.2
Terbuthylazine	0.03	0.2
Metamitron	0.05	0.5
Metazachlor	0.01	0.2
Metolachlor	0.01	0.1
Ethofumesate	0.10	0.2

models: (a) single porosity (SP) et (b) dual porosity (DP) (Šimůnek et al., 2013; Šimůnek et al., 2008; van Genuchten and Wierenga, 1976). In both cases, a modified form of Richard's equation is solved.

2.6.2.1. Single-porosity model (SP). For the single porosity model, the soil water retention curve $\theta(h)$ is described by the van Genuchten (1980) equation and the hydraulic conductivity function $K(h)$ is described with the statistical pore distribution model of Mualem (1976) with:

$$Se = \frac{\theta - \theta_r}{\theta_s - \theta_r}, m = 1 - \frac{1}{n} \quad n > 1 \quad (2)$$

where S_e is the effective saturation (—), θ is the soil water content (L^3/L^3), θ_r and θ_s are the residual and saturation water contents (L^3/L^3) respectively and n and m are hydraulic parameters (—).

The five independent soil hydraulic parameters required are θ_r , θ_s , n , the hydraulic shape parameter α ($1/L$), the saturated hydraulic conductivity K_s (L/T) and the pore connectivity l (—). The l parameter has been estimated to be 0.5 on average for most soils (Köhne et al., 2006; Mualem, 1976). These parameters were first estimated using neural network based pedotransfer functions developed by Schaap et al. (2001) and incorporated in the Rosetta module (Table 5). The granulometry and bulk density of the soil columns (Table 1) were used. θ_s is considered to be the total porosity of the columns. θ_r has only a minor influence on water fluxes because the soil columns retain a quite high water content for the duration of the experiment. In addition, this value is generally kept constant using estimates of pedotransfer functions (Mertens et al., 2009; Sur et al., 2022). Only K_s has been calibrated using the cumulative outflow volumes of the columns. Mertens et al., 2006 showed that saturated hydraulic conductivity (K_s) was the most sensitive parameter when inverse modelling was performed using eluted water volume data.

2.6.2.2. Dual-porosity model (DP). The DP approach considers the porous media to be divided into two interacting regions, partitioning the porous system into mobile and immobile regions. DP models assume that water flow is confined to the macropore or fracture domain. Water in the matrix does not move at all (Šimůnek et al., 2003; van Genuchten and Wierenga, 1976). Water can be exchanged between the two regions. We will use the subscript *mo* for the mobile or macropore region and *im* for the immobile region or soil matrix.

Total water content (L^3/L^3) is given by:

$$\theta = \theta_{mo} + \theta_{im} \quad (3)$$

The DP model uses the Richards equation to describe water flow in the mobile region (macropores). A mass balance equation is used to describe water dynamics in the matrix (Šimůnek et al., 2003):

Table 5

Initial values of hydraulic parameters determined by the Rosetta module from granulometry and bulk density. The values of θ_s (third column) correspond to the total porosity of the columns, calculated from bulk density and organic matter content.

	θ_r (cm^3/cm^3)	θ_s (cm^3/cm^3)	α ($1/cm$)	n (—)	K_s (cm/day)
RH1	0.065 ±	0.43 ±	0.0054 ±	1.66 ±	24.6 ± 2.5
	0.001	0.01	0.0001	0.01	
	0.061 ±	0.40 ±	0.0063 ±	1.61 ±	
RH2	0.001	0.00	0.0001	0.01	14.3 ± 0.8
	0.065 ±	0.39 ±	0.0068 ±	1.56 ±	
	0.000	0.00	0.0001	0.00	
RH3	0.055 ±	0.38 ±	0.0071 ±	1.57 ±	5.9 ± 0.2
	0.000	0.00	0.0001	0.01	
	0.065 ±	0.40 ±	0.0064 ±	1.59 ±	
IH1	0.000	0.00	0.0001	0.01	10.6 ± 0.5
	0.065 ±	0.40 ±	0.0064 ±	1.59 ±	
VH1	0.002	0.01	0.0003	0.02	8.9 ± 1.5

$$\frac{\partial \theta_{mo}}{\partial t} = \frac{\partial}{\partial x} \left[K(h) \left(\frac{\partial h}{\partial x} + \cos \alpha \right) \right] - S_{mo} - \Gamma_w \quad (4)$$

$$\frac{\partial \theta_{im}}{\partial t} = -S_{im} + \Gamma_w \quad (5)$$

where S_{mo} and S_{im} are the sink terms for the two regions (1/T) and Γ_w is the rate of water transfer from the macropores to the matrix and vice versa (1/T). In the current version of Hydrus 1-D (4.17), the S_{im} term is assumed to be zero.

The mass transfer rate was assumed to be proportional to the difference in the effective saturations of the two regions using the first order rate equation (Gerke and van Genuchten, 1993; Šimůnek et al., 2001):

$$\Gamma_w = \frac{\partial \theta_{im}}{\partial t} = \omega [S_{e_{mo}} - S_{e_{im}}] \quad (6)$$

where ω is the first-order rate coefficient for water transfer between mobile and immobile regions (1/T) and $S_{e_{mo}}$ and $S_{e_{im}}$ are the effective fluid saturations of the mobile (macropores) and immobile (matrix) regions.

Hydrus 1-D assumes that the retention curves of both regions are similar. Therefore, the parameters α and n in the van Genuchten equation have the same value. For the mobile region, the parameters required for the van Genuchten-Mualem model are $\theta_{r,mo}$, $\theta_{s,mo}$, α and n and K_s . For the immobile region the required parameters are $\theta_{r,im}$, $\theta_{s,im}$ et ω . The parameters α , n and K_s are identical to those used in the SP modelling.

The parameters $\theta_{s,mo}$ and $\theta_{s,im}$ are calculated taking into account a proportion of macropores (mobile region) of 10 % of the total porosity (Kodešová et al., 2005; Šimůnek et al., 2009). $\theta_{r,im}$ is considered equal to θ_r , from the SP model (Imig et al., 2023b). $\theta_{r,mo}$ is set to zero so that residual water is only present in the immobile region (Imig et al., 2023b; Köhne et al., 2006; Köhne et al., 2002). The first-order rate coefficient for water transfer ω was set to 0.2 day⁻¹ (Köhne et al., 2004). The parameters $\theta_{s,mo}$ and ω were optimised by inverse modelling for each column. Then $\theta_{s,im}$ was calculated according to the total water content equation. The other parameters were kept constant.

2.6.3. Pesticides transport modelling

Only the DP model was used to model pesticide fluxes. The transport of solutes is described by the convection-dispersion equation. The exchange of solutes between the two liquid regions can be modelled as a first order process. Assuming that degradation is only active in the liquid phase, the governing equation is written as follows:

$$\frac{\partial \theta_{mo} c_{mo}}{\partial t} + f_{mo} \rho \frac{\partial s_{mo}}{\partial t} = \frac{\partial}{\partial x} \left(\theta_{mo} D_{mo} \frac{\partial c_{mo}}{\partial x} \right) - \frac{\partial q_{mo} c_{mo}}{\partial x} - \mu \theta_{mo} c_{mo} - \Gamma_s \quad (7)$$

$$\frac{\partial \theta_{im} c_{im}}{\partial t} + \left(1 - f_{mo} \right) \rho \frac{\partial s_{im}}{\partial t} = \Gamma_s - \mu \theta_{im} c_{im} \quad (8)$$

$$\Gamma_s = \omega_s (c_{mo} - c_{im}) + \Gamma_w c^* \quad (9)$$

In witch, c_{mo} and c_{im} are the solute liquid concentrations of the mobile and immobile regions (ML-3), s_{mo} and s_{im} are the sorbed concentrations of the mobile and immobile regions (–), D_{mo} is the dispersion coefficient in the mobile region, taking into account molecular diffusion and hydrodynamic dispersion (L²/T), q_{mo} is the volumetric flux density in the mobile region (L/T), f_{mo} is the fraction of sorption sites in contact with the mobile region (–), Γ_s is the solute transfer rate between the two regions (M/L³T) and μ (1/T) is the first order degradation rate coefficient in the liquid phase. ω_s is the first order chemical mass transfer coefficient (1/T) and c^* is the solute concentration (M/L³) of the mobile region c_{mo} for $\Gamma_w > 0$ and of the immobile region c_{im} for $\Gamma_w < 0$ (Šimůnek and Genuchten, 2008).

Chemical equilibrium with Freundlich sorption isotherms are used for both regions:

$$s_{mo} = K_f c_{mo}^\beta$$

$$s_{im} = K_f c_{im}^\beta \quad (10)$$

with K_f the Freundlich adsorption isotherm coefficient (L³/M) and β the Freundlich exponent 1/n (–).

The dispersion coefficient of the mobile zone D_{mo} can be described as:

$$D_{mo} = D_0 \frac{\lambda}{\theta_{s,mo}^2} + \lambda_{mo} \nu \quad (11)$$

where D_0 is the molecular diffusion coefficient in free water (L²/T), λ_{mo} is the longitudinal dispersivity in the mobile region (L) and ν is the pore water velocity (L/T). In general, the molecular diffusion coefficient is considered negligible due to its small contribution to the dispersion (Šimůnek et al., 2009). Indeed, the molecular diffusion coefficient is generally of the order of 5 % of the hydrodynamic dispersion coefficient (Vanderborght and Vereecken, 2007).

The dispersivity λ , the degradation rate coefficient μ , the Freundlich adsorption coefficient K_f , the Freundlich coefficient β and the first order chemical mass transfer coefficient ω_s are required. The initial value of the transfer coefficient ω_s has been set to 0.02 day⁻¹ (Šimůnek et al., 2009).

For λ , μ , K_f and β , the optimisation was performed from three initial values. In fact, local algorithms such as Levenberg-Marquardt will find a local optimum from the initial parameters entered into the model. Thus, the optimised parameters will depend on the initial values (Dubus et al., 2004). Therefore, the mean values obtained from the optimisation using three initial parameter values will be more robust. For the dispersivity, initial values of 1, 3 and 6 cm were used. This parameter was only estimated for bentazone and then kept constant for the other pesticides. This is because bentazone shows very low adsorption and can be assimilated to a tracer as observed by De Wilde et al. (2009). There are several types of tracer, with different adsorption and degradation characteristics, each suitable for different purposes.

For the eight pesticides, the initial values of μ , K_f and β were taken from the peer review of the pesticide risk assessment provided by the European Food Safety Authority (EFSA). EFSA provides conclusions to these peer reviews at the request of the European Commission. The K_f , β and half-life DT_{50} (T) values of the studies considered in the peer review, together with the corresponding country and soil type, are given in the annex 1. The Walloon Agronomical Research Centre (CRA–W) has compiled from EFSA, the values measured in the same climatic zone and for silty soils such as those found in Wallonia, Belgium. The minimum, mean and maximum values of the selected data were used as initial pesticide values for the different parameters (Table 6). The field DT_{50} were transformed into the degradation rate coefficient μ using the equation: $\mu = \ln(2)/DT_{50}$. The parameters K_f , β and μ were optimised with the three sets of input values for the fifteen columns and eight pesticides.

2.6.4. Boundary conditions

For the numerical modelling with Hydrus 1-D, 30 cm columns were represented as 1D domains and discretised into 101 nodes with equal spacing of 0.3 cm. For the water flow boundary conditions, the top of the columns was set to an atmospheric boundary condition with a surface water layer to allow for water build-up during time variable boundary conditions of water addition (precipitation). At the bottom of the column, the boundary condition was set to a constant pressure head due to the vacuum pump. An initial pressure head of –510 cm was applied at the bottom of the column by the pump. The initial head at the top of the column was determined for each column as the pressure at field capacity, the columns having been saturated and drained for 24 h prior to the start of the experiment. For the solute transport, a boundary

Table 6

Initial values of the transport parameters of the pesticides obtained from the EFSA peer reviews. The initial values are chosen as the minimum, mean and maximum of the values retained by the CRA-W for the Walloon soil-climate zone.

	K_f (cm ³ /g)			β (–)			Field DT ₅₀ (days)		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
Bentazone	0.08	0.76	1.94	0.70	0.97	1.00	3.0	4.7	6.9
Metamitron	0.56	3.06	4.65	0.65	0.77	0.85	6.6	11.1	21.7
Metazachlor	0.48	2.75	3.50	0.82	0.84	0.85	5.1	8.50	14.4
S-Metolachlor	0.59	5.33	4.80	0.91	0.93	0.93	3.6	22.4	55.7
Ethofumesate	0.70	2.95	6.20	0.82	0.90	0.96	13.1	40.8	110.0
Flufenacet	1.00	5.05	14.7	0.84	0.87	0.98	13.1	27.7	54.2
Terbuthylazine	2.10	5.78	10.5	0.88	0.90	0.98	6.9	19.5	35.7
Chlorotoluron	2.70	4.52	9.60	0.82	0.87	0.93	26.0	42.0	42.0

condition flux concentration was chosen for the top of the columns. This corresponds to the pesticide pulse at the beginning of the experiment. The lower boundary condition of the columns was set to zero concentration gradient.

2.7. Data treatment and statistical analysis

Breakthrough curves were constructed by plotting the relative concentrations (C/C₀) measured in the leachates against the cumulative relative pore volume (V/V₀). The concentrations of pesticides found in the eluates (C) are divided by the concentrations of the pesticide pulse initially applied to the columns (C₀) to obtain the relative concentrations. Based on the porosity of each column, the total pore volume (V₀) was determined from the volume of the soil column. The volume of water for each sample (V) is then divided by the total pore volume (V₀) to obtain the relative pore volume. This type of data allows different columns with different porosities, soil volumes and pesticides pulse concentrations to be compared. Data were processed using Rstudio v2023.12.0. The figures were plotted with ggplot2 package. The T_{min}, C_{peak} and T_{peak} indicators were determined as the relative pore volume at the first appearance of the pesticide in the eluates, the maximum relative concentration and the relative pore volume after which it was reached, for each pesticide and column.

Analyses of variance (ANOVA) were performed to examine differences in parameters between pesticides, depths and agricultural practices over time. ANOVA tests are used to identify differences between the means of more than two groups and to establish whether these differences are due to chance or whether they reflect a meaningful difference. This test assumes that the data are normal and that the variances are equal between the groups. Normality was tested using Shapiro-Wilk and equality of variances using Levene's test. All assumptions were satisfied. ANOVA tests are regularly used to compare several factors between different soil depths and agricultural practices (Alavaisha et al., 2019; de Paul Obade and Lal, 2014; Leul et al., 2023). Fisher's Least Significant Difference (LSD) method was used to compare means with the null hypothesis of equality of means and a significance level of 0.05.

3. Results

3.1. Breakthrough curves analysis

Fig. 2 displays the measured and modelled breakthrough curves (BTCs) of the eight pesticides for the three agricultural systems (RH1, IH1 and VH1) and the three soil horizons (RH1, RH2, RH3). BTCs are plotted as the measured relative concentration (C/C₀) versus the number of pore volume eluted (V/V₀). The points correspond to the measured data and the lines to the modelled data.

3.1.1. Breakthrough curves indicators

The arrival times of the first breakthrough T_{min}, the arrival time of the maximal peak concentration T_{peak} and the relative maximum concentrations of the peak C_{peak} for the eight pesticides, the three

production systems and the three depths are given in Table 7.

Pesticide breakthrough curves show varied patterns between pesticides, cropping systems and depths. All pesticides are found faster in the leachates with a significantly lower time of the first breakthrough (T_{min}) for horizons 2 and 3 (RH2 and RH3) than for horizon 1 (RH1), where they are more retained. Bentazone, S-metolachlor, flufenacet and terbuthylazine were eluted faster in the vegan system (VH1) with a lower T_{min} than in the reference system (RH1) and in the ICLS system (IH1). For the others, no significant difference was found between the cropping systems. The peak appeared later in the topsoil, with a significantly higher arrival time of the maximal peak concentration T_{peak} for RH1 than for the deeper horizons RH2 and RH3, indicating greater retention of pesticides in the surface. For the three cropping systems, T_{peak} was significantly higher for RH1 and IH1 than for VH1 for chlorotoluron, flufenacet and terbuthylazine. Ethofumesate was the exception, with a higher T_{peak} for IH1 than for RH1 and VH1. No significant difference was found for the others. For all pesticides except bentazone, where no difference was observed, the maximal concentration of the peak (C_{peak}) was significantly higher for RH2 and RH3 than for RH1. No significant difference in C_{peak} was observed between the systems, except for bentazone, where C_{peak} of IH1 was higher than that of VH1.

Bentazone is the most rapidly displaced pesticide in all columns and therefore the first compound to appear in the leachates. Its T_{min} and T_{peak} are lower and its C_{peak} greater than those of the other pesticides. These differences are significant for RH1 and IH1 for T_{min} and T_{peak} and for all columns for C_{peak}. T_{min} was between 0.01 and 0.03 and T_{peak} between 0.15 and 0.45. This means that the peak is reached after less than 0.5 pore volumes eluted, indicating non-equilibrium transport. The bentazone BTCs are strongly skewed to the left with a long tail, indicating a high degree of preferential flow within the columns. Small differences in the degree of preferential flow were observed between the production systems. However, the C_{peak} of IH1 (0.410 ± 0.029) is significantly higher than that of VH1 (0.229 ± 0.071) for a lower of T_{peak}. Between horizons, T_{peak} is significantly lower for RH2 than for RH1, confirming a greater degree of preferential flow within horizon 2. After bentazone, metazachlor and metamitron generally appeared in the leachates before the other pesticides. They have a lower T_{peak} and a higher C_{peak}, indicating a low retention within the columns. However, these two pesticides are delayed compared to bentazone. In general, no significant differences in T_{peak} and C_{peak} are observed between these two pesticides, showing a similar leaching pattern. Their peaks occur after 0.44–2.14 and 0.37–2.14 pore volume for metazachlor and metamitron, respectively, for C_{peak} between 0.020 and 0.132 and 0.016–0.102. In contrast, terbuthylazine and flufenacet are the most strongly retained within the columns. T_{peak} ranged from 1.46 to 6.26 and 1.72–4.99 pore volume, with C_{peak} values of 0.006–0.042 and 0.002–0.024 for terbuthylazine and flufenacet, respectively. Their BTCs are wider and flatter than for the other pesticides, with higher T_{peak} and lower C_{peak} for the majority of columns.

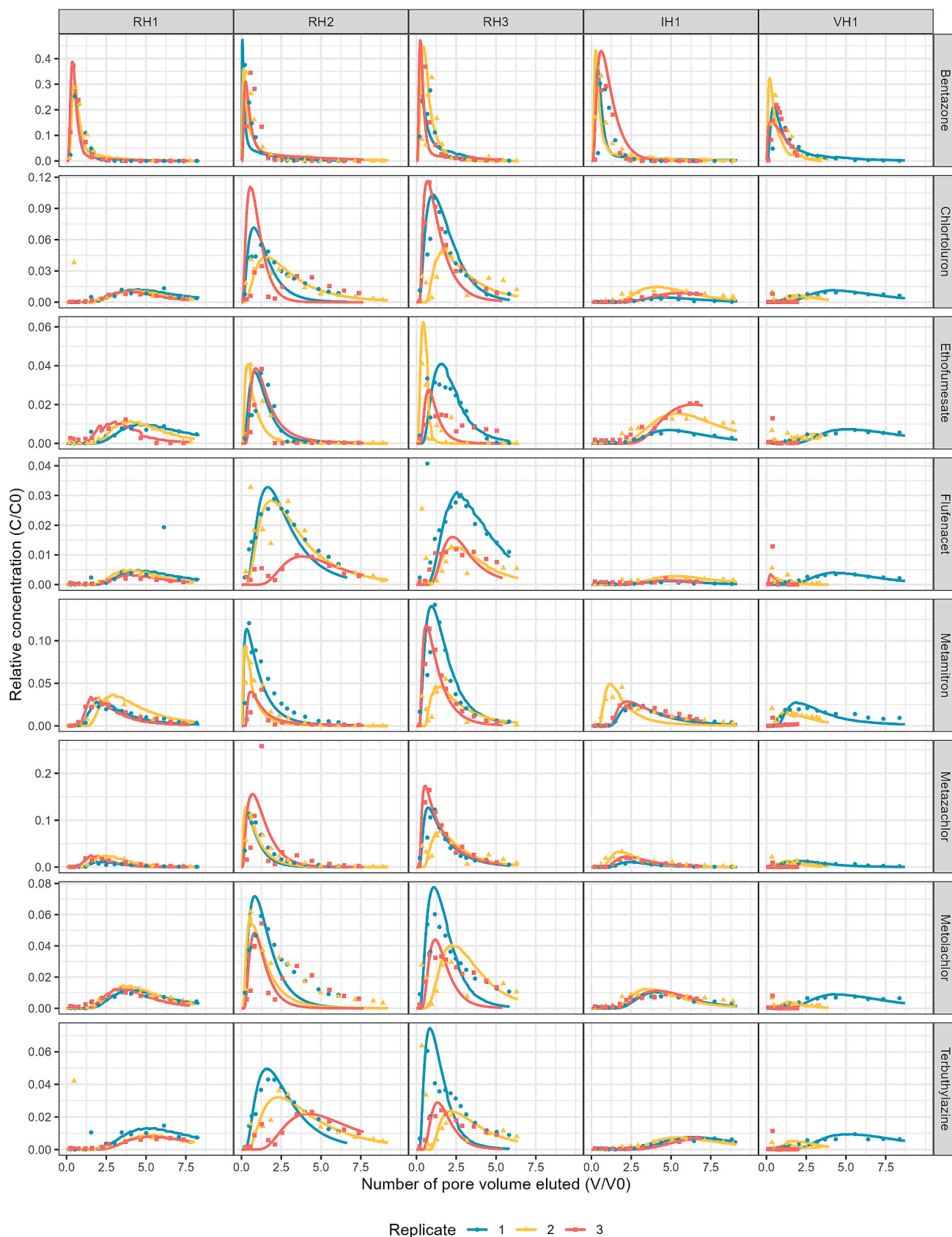


Fig. 2. Measured and modelled breakthrough curves depicting the relative concentration (C/C_0) as a function of the number of pore volume eluted (V/V_0) for the eight herbicides, the three agricultural systems and the three horizons. The three spatial replicates are plotted. The points correspond to the measured data and the lines to the modelled data.

Table 7

Arrival time of the first breakthrough T_{min} in pore volume, arrival time of the maximum peak concentration T_{peak} in pore volume and relative maximal concentration of the peak C_{peak} for the eight pesticides, the three production systems and the three depths. Values correspond to the mean of the three spatial replicates with standard deviation.

Pesticide	Column	T_{min} [–]			T_{peak} [–]			C_{peak} [–]		
Bentazone	RH1	0,03	±	0,01	0,45	±	0,06	0,318	±	0,052
	RH2	0,01	±	0,00	0,15	±	0,08	0,381	±	0,068
	RH3	0,01	±	0,00	0,32	±	0,08	0,391	±	0,096
	VH1	0,01	±	0,00	0,31	±	0,11	0,229	±	0,071
	IH1	0,02	±	0,00	0,42	±	0,14	0,410	±	0,029
Chlorotoluron	RH1	0,96	±	0,16	4,22	±	0,22	0,011	±	0,001
	RH2	0,06	±	0,04	0,95	±	0,42	0,075	±	0,028
	RH3	0,26	±	0,20	1,10	±	0,37	0,089	±	0,029
	VH1	0,38	±	0,16	3,09	±	1,17	0,009	±	0,003
	IH1	0,97	±	0,18	4,76	±	0,60	0,009	±	0,004
Ethofumesate	RH1	1,00	±	0,21	3,88	±	0,61	0,011	±	0,001
	RH2	0,11	±	0,08	0,71	±	0,21	0,039	±	0,002
	RH3	0,15	±	0,09	0,92	±	0,48	0,044	±	0,014
	VH1	1,21	±	0,06	3,93	±	0,97	0,006	±	0,001
	IH1	1,37	±	0,43	5,53	±	0,63	0,014	±	0,005
Flufenacet	RH1	1,29	±	0,07	4,22	±	0,22	0,004	±	0,001
	RH2	0,35	±	0,26	2,44	±	0,97	0,024	±	0,010
	RH3	0,55	±	0,11	2,35	±	0,14	0,020	±	0,008
	VH1	0,37	±	0,36	1,72	±	1,81	0,003	±	0,001
	IH1	1,12	±	0,11	4,99	±	0,22	0,002	±	0,001
Metamitron	RH1	0,90	±	0,20	2,14	±	0,63	0,033	±	0,004
	RH2	0,11	±	0,09	0,37	±	0,13	0,083	±	0,031
	RH3	0,26	±	0,15	0,92	±	0,24	0,102	±	0,040
	VH1	0,50	±	0,10	1,63	±	0,23	0,016	±	0,011
	IH1	0,86	±	0,30	2,01	±	0,59	0,035	±	0,011
Metazachlor	RH1	0,45	±	0,17	2,14	±	0,43	0,020	±	0,006
	RH2	0,02	±	0,00	0,44	±	0,18	0,132	±	0,018
	RH3	0,20	±	0,11	0,94	±	0,44	0,125	±	0,040
	VH1	0,37	±	0,13	1,64	±	0,26	0,013	±	0,000
	IH1	0,65	±	0,12	2,10	±	0,28	0,022	±	0,009
S-Metolachlor	RH1	0,85	±	0,07	3,70	±	0,21	0,013	±	0,001
	RH2	0,06	±	0,02	0,73	±	0,13	0,060	±	0,010
	RH3	0,25	±	0,10	1,49	±	0,51	0,054	±	0,017
	VH1	0,43	±	0,26	2,88	±	1,38	0,007	±	0,003
	IH1	0,88	±	0,10	3,87	±	0,35	0,011	±	0,001
Terbuthylazine	RH1	1,22	±	0,12	5,42	±	0,26	0,010	±	0,002
	RH2	0,32	±	0,24	2,70	±	1,10	0,035	±	0,011
	RH3	0,40	±	0,28	1,46	±	0,57	0,042	±	0,023
	VH1	0,51	±	0,40	2,55	±	1,69	0,006	±	0,003
	IH1	1,60	±	0,13	6,26	±	0,64	0,007	±	0,000

3.1.2. Pesticides mass balance

Fig. 3 shows the pesticide mass balances with measured data on the percentages recovered from the leachates and remaining adsorbed in the columns, as well as the resulting degraded percentages. The behaviour of pesticides in terms of elution, adsorption and degradation varies considerably depending on the pesticide, cropping system and horizons.

The recovery rate of pesticides in the leachates was higher for RH2 and RH3 than for RH1, except for bentazone, metamitron and ethofumesate, which showed no difference. The amounts of pesticides recovered in the leachates were similar between the three cropping systems for all pesticides. Recovery rates varied considerably between pesticides. For bentazone, higher amounts were recovered than for the other pesticides in the three cropping systems, ranging from 73.4 to 81.3 % of the applied amount. The second highest amount recovered in the leachates is that of metamitron, ranging from 15.5 to 33.1 %. There was no significant difference between the other pesticides, which were all below 20 % recovery. The pesticide with the lowest recovery from the topsoil

was flufenacet for all three cropping systems, with recoveries ranging from 2.5 to 7.2 %.

For RH2 and RH3, bentazone and metazachlor had a significantly higher recovery rate than the others, with 77.13–81.16 % and 59.8–64.3 % of the initial applied amount, respectively. The two pesticides with the lowest recoveries were flufenacet, as in the topsoil, with between 20.1 and 27.0 %, and ethofumesate in last place, with between 12.6 and 16.7 %.

For metazachlor, S-metolachlor, flufenacet, chlorotoluron and terbuthylazine, a significantly higher amount of pesticides was degraded in RH1 than in RH2 and RH3. Degradation was similar between horizons for the other pesticides and also between cropping systems for all pesticides. Within the topsoil, flufenacet and metazachlor were the most degraded pesticides with between 89.8 and 95.0 and 82.2–87.6 % respectively. Bentazone had a significantly lower degradation percentage than the other pesticides, ranging between 18.1 and 26.7 %. Metamitron was the second least degraded pesticide, ranging from 61.3 to

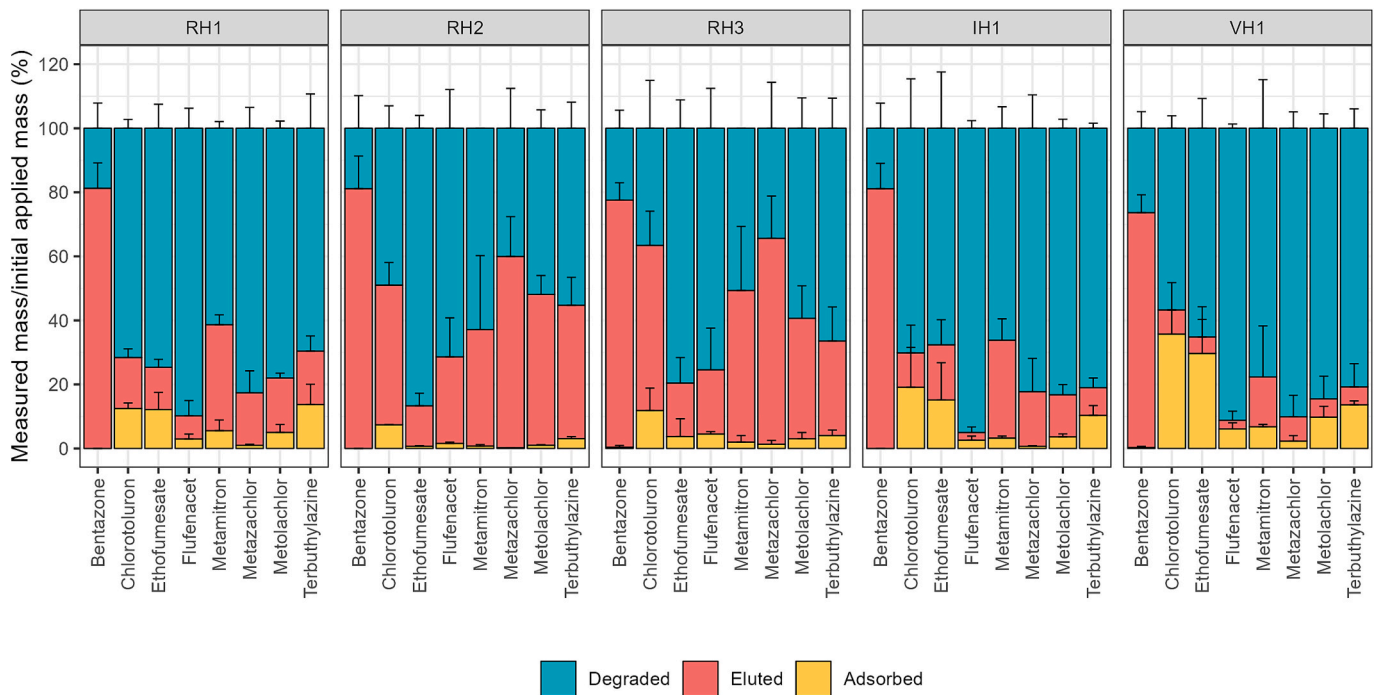


Fig. 3. Pesticide mass balances with the percentage recovered in the leachates, degraded and adsorbed within the columns for the eight pesticides, the three cropping systems and the three horizons. The error bars represent the standard deviation.

77.6 %, but was not significantly different from the more degraded pesticides. The pesticides showed different degradation behaviour in the deeper horizons compared to the topsoil horizon. For RH2 and RH3, ethofumesate and flufenacet are significantly more degraded than the other pesticides with values between 79.5 and 86.7 and 71.4–75.4 % respectively. The two least degraded pesticides are metazachlor and bentazone with between 34.4 and 40.0 and 18.8–22.5 % respectively, with no significant difference between the two. Thus, except for bentazone, metazachlor changed from the most degraded pesticide in horizon 1 to the least degraded pesticide in horizons 2 and 3. The opposite trend is observed for ethofumesate, which goes from being one of the least degraded pesticides in horizon 1 to the most degraded in horizons 2 and 3.

The pesticide adsorption percentages for the different cropping systems and horizons exhibit greater variability. Except for bentazone, metazachlor and terbutylazine, the pesticides show higher adsorption in VH1 than in RH1 and IH1. For RH1 and IH1, the most adsorbed pesticides were chlorotoluron, ethofumesate and terbutylazine with 12.5–19.11 %, 12.1–15.4 % and 10.3–13.7 % respectively. For VH1, only chlorotoluron and ethofumesate differed significantly from the other pesticides with 31.6 and 29.7 % adsorption. The two least adsorbed pesticides, which did not differ significantly from each other or from the cropping systems, were bentazone and metazachlor with 0.0–0.3 and 3.4–6.8 %, respectively. For metamitron, S-metolachlor and terbutylazine, a higher percentage of adsorption was measured in RH1 than in RH2 and RH3. For metazachlor and flufenacet, greater adsorption was measured at 60–90 cm depth (RH3) than at 30–60 cm (RH2). Finally, bentazone, chlorotoluron and ethofumesate showed no significant differences in adsorption between the three horizons. Within RH2 and RH3, chlorotoluron was significantly more adsorbed than the other pesticides, ranging from 7.4 to 11.9 %. Terbutylazine and flufenacet followed with 3.1–4.0 and 1.6–4.6 % respectively. However, their adsorption percentage was significantly higher than the other pesticides only for RH2. Finally, bentazone and metazachlor were also the least adsorbed pesticides with between 0.0 and 0.4 and 0.2–1.3 %, respectively.

3.2. Inverse modelling parameters

3.2.1. Soil hydraulics parameters

The hydraulic parameters obtained by inverse modelling of the cumulative outflows on Hydrus 1-D for all columns are given in Table 8.

Good agreement was achieved between observed and predicted cumulative outflow. The R^2 is between 0.98 and 1.00 and the RMSE between 0.02 and 0.05 cm of water for all columns. The VH1 columns showed a weaker fit due to the third replicate. In fact, unlike the other two, water did not flow properly and remained stagnant on the column surface. The K_s for this column is only 0.20 cm/day for a total water outflow of 23.73 cm.

Table 8

Soil hydraulics parameters obtained by inverse modelling of cumulative outflow for the eight pesticides, the three cropping systems and the three depths. Values correspond to the mean value of the three spatial replicates with the standard deviation.

	$\theta_{s,mo}^b$ (cm ³ /cm ³)	$\theta_{s,im}^c$ (cm ³ /cm ³)	Proportion of the mobile zone ^c (%)	K_s^a (cm/day)	ω^b (1/day)	R^2 (–)	RMSE (cm)
RH1	0.036	0.43 ± 0.01	7.8 ± 1.5	3.33	0.26	1.00	0.02
	±			±	±	±	± 0.00
	0.007			0.14	0.04	0.00	
	0.039			0.89	0.22	1.00	0.02
RH2	±	0.38 ± 0.01	9.5 ± 2.1	±	±	±	± 0.01
	0.009			0.15	0.01	0.00	
	0.028			0.94	0.17	0.99	0.03
	±			±	±	±	± 0.00
RH3	0.007	0.35 ± 0.01	7.4 ± 2.0	0.21	0.02	0.00	
	0.038			0.94	0.20	1.00	0.02
	±			±	±	±	± 0.00
	0.003			0.18	0.00	0.00	
IH1	0.035	0.34 ± 0.01	10 ± 0.8	0.69	0.20	0.98	0.05
	±			±	±	±	± 0.02
	0.003			0.46	0.00	0.01	
	0.003			0.46	0.00	0.01	
VH1	±	0.36 ± 0.01	8.8 ± 0.9	±	±	±	± 0.02
	0.003			0.46	0.00	0.01	
	0.003			0.46	0.00	0.01	
	0.003			0.46	0.00	0.01	

^a Parameter determined by equilibrium modelling.

^b Parameters determined by DP modelling.

^c Parameters calculated from the estimated parameters.

Total cumulative outflow was significantly higher for columns RH1 and IH1 with 107.8 ± 3.5 cm and 91.4 ± 15.1 cm respectively than for VH1 with 54.0 ± 37.2 cm. The cumulative outflow was also higher for RH1 than for RH3 with 64.8 ± 6.86 cm. The K_s was significantly higher for RH1 than for the other column types. Thus, in the other production systems with maize (IH1) and camelina (VH1), as well as in the deeper horizons (RH2 and RH3), water infiltration is slower at saturation. The percentage of mobile zone shows no significant differences. Significant differences were observed between the bulk densities (BD) of the columns. IH1 has the highest BD, followed by VH1 and finally RH1. At horizon level, the BDs were ranked with depth, with RH3, then RH2 and finally RH1.

3.2.2. Pesticides transport parameters

Transport parameters for the eight pesticides obtained by inverse modelling leachate concentrations are presented in Table 9. The BTCs for the eight pesticides are well described by the dual porosity (DP) modelling, as shown by the high R^2 values, above 0.7 for most of the

columns. The VH1 columns showed slightly lower R^2 values. The third replicate could only be modelled for bentazone and metamitron. For the other pesticides, no acceptable fit was obtained due to very poor water flows within this column, leading to erratic BTCs.

The dispersivity (λ) was estimated for bentazone because of its behaviour as a tracer with very low adsorption. The values were then kept constant for the other pesticides. Columns VH1 showed significantly greater dispersivity than columns IH1 and RH1, which remained close to the initial value of 3 cm. For the three horizons, a greater dispersivity was obtained for the columns from horizon 2, RH2, than for those from horizons 1 and 3 (RH1 and RH3).

The pesticide transport parameters, the Freundlich adsorption coefficient K_f , the first order degradation rate coefficient μ or the half-life DT_{50} and the first order chemical mass transfer coefficient ω_s were highly variable between columns for each pesticide.

For the horizons, the K_f of horizon 1 was always significantly higher than those of horizons 2 and/or 3 for all pesticides. For most pesticides, no significant difference was observed between the K_f for the different

Table 9

Fitted pesticide transport parameters determined by inverse modelling of leachate concentrations on Hydrus 1-D for the eight pesticides, the three cropping systems and the three horizons. Values correspond to the mean value of the three spatial replicates with the standard deviation.

Pesticide	Column	λ_{mo} [cm]	μ [1/day]	K_f [cm ³ /g]	β [-]	ω_s [cm ³ /g]	R^2 [-]	RMSE [mg/cm ³]
Bentazone	RH1	2.87 ± 0.21	0.04 ± 0.02	0.22 ± 0.02	1.00 ± 0.00	0.01 ± 0.01	0.98 ± 0.01	0.04 ± 0.00
	RH2	7.02 ± 2.23	0.02 ± 0.02	0.10 ± 0.06	1.00 ± 0.00	0.02 ± 0.00	0.95 ± 0.02	0.07 ± 0.03
	RH3	4.16 ± 1.15	0.07 ± 0.06	0.15 ± 0.03	1.00 ± 0.00	0.02 ± 0.01	0.90 ± 0.06	0.09 ± 0.02
	IH1	3.00 ± 0.63	0.06 ± 0.06	0.17 ± 0.06	1.00 ± 0.00	0.03 ± 0.03	0.93 ± 0.04	0.11 ± 0.05
	VH1	5.92 ± 1.00	0.01 ± 0.00	0.17 ± 0.03	1.00 ± 0.00	0.02 ± 0.01	0.94 ± 0.01	0.09 ± 0.04
Chlorotoluron	RH1	–	0.50 ± 0.04	1.88 ± 0.08	0.92 ± 0.02	0.57 ± 0.08	0.87 ± 0.08	0.12 ± 0.02
	RH2	–	0.20 ± 0.02	0.50 ± 0.31	0.93 ± 0.03	0.21 ± 0.11	0.76 ± 0.12	0.18 ± 0.06
	RH3	–	0.15 ± 0.08	0.42 ± 0.19	0.82 ± 0.09	0.11 ± 0.02	0.83 ± 0.20	0.13 ± 0.09
	IH1	–	0.63 ± 0.16	1.78 ± 0.54	0.94 ± 0.01	0.57 ± 0.23	0.85 ± 0.07	0.15 ± 0.04
	VH1	–	0.50 ± 0.00	1.57 ± 0.31	0.94 ± 0.01	0.34 ± 0.00	0.68 ± 0.13	0.25 ± 0.08
Ethofumesate	RH1	–	0.48 ± 0.02	1.59 ± 0.36	0.90 ± 0.00	2.04 ± 1.2	0.82 ± 0.10	0.12 ± 0.05
	RH2	–	0.53 ± 0.16	0.36 ± 0.06	0.84 ± 0.05	0.40 ± 0.21	0.83 ± 0.06	0.12 ± 0.03
	RH3	–	0.53 ± 0.18	0.33 ± 0.20	0.89 ± 0.03	0.37 ± 0.14	0.61 ± 0.09	0.25 ± 0.02
	IH1	–	0.49 ± 0.15	1.67 ± 0.32	0.91 ± 0.05	0.34 ± 0.10	0.89 ± 0.06	0.12 ± 0.03
	VH1	–	0.60 ± 0.03	1.29 ± 0.49	0.76 ± 0.09	0.27 ± 0.14	0.56 ± 0.26	0.36 ± 0.21
Flufenacet	RH1	–	0.89 ± 0.06	1.66 ± 0.16	0.89 ± 0.01	0.90 ± 0.15	0.94 ± 0.02	0.11 ± 0.05
	RH2	–	0.33 ± 0.06	1.07 ± 0.26	0.90 ± 0.00	0.61 ± 0.58	0.71 ± 0.08	0.22 ± 0.04
	RH3	–	0.45 ± 0.22	0.77 ± 0.14	0.87 ± 0.03	0.26 ± 0.02	0.66 ± 0.14	0.23 ± 0.07
	IH1	–	1.38 ± 0.17	2.01 ± 0.28	0.95 ± 0.02	0.74 ± 0.34	0.57 ± 0.11	0.30 ± 0.06
	VH1	–	1.13 ± 0.01	1.19 ± 0.63	0.89 ± 0.02	0.60 ± 0.16	0.66 ± 0.23	0.19 ± 0.06
Metamitron	RH1	–	0.35 ± 0.07	1.15 ± 0.14	0.75 ± 0.02	0.14 ± 0.05	0.93 ± 0.04	0.08 ± 0.03
	RH2	–	0.29 ± 0.12	0.26 ± 0.05	0.62 ± 0.06	0.16 ± 0.05	0.87 ± 0.09	0.11 ± 0.03
	RH3	–	0.17 ± 0.12	0.44 ± 0.21	0.74 ± 0.04	0.24 ± 0.15	0.83 ± 0.22	0.09 ± 0.08
	IH1	–	0.42 ± 0.11	0.86 ± 0.34	0.67 ± 0.09	0.18 ± 0.04	0.86 ± 0.08	0.13 ± 0.05
	VH1	–	0.43 ± 0.00	0.83 ± 0.03	0.72 ± 0.02	0.12 ± 0.03	0.66 ± 0.09	0.29 ± 0.06
Metazachlor	RH1	–	0.59 ± 0.22	0.94 ± 0.14	0.92 ± 0.05	0.54 ± 0.04	0.88 ± 0.07	0.11 ± 0.04
	RH2	–	0.14 ± 0.05	0.18 ± 0.01	0.87 ± 0.04	0.16 ± 0.05	0.85 ± 0.15	0.10 ± 0.06
	RH3	–	0.28 ± 0.31	0.35 ± 0.16	0.84 ± 0.01	0.03 ± 0.00	0.92 ± 0.09	0.08 ± 0.05
	IH1	–	0.59 ± 0.18	0.64 ± 0.15	0.83 ± 0.01	0.38 ± 0.16	0.91 ± 0.08	0.10 ± 0.04
	VH1	–	0.58 ± 0.16	0.69 ± 0.07	0.84 ± 0.00	0.29 ± 0.05	0.64 ± 0.08	0.23 ± 0.01
S-Metolachlor	RH1	–	0.49 ± 0.01	1.71 ± 0.12	0.92 ± 0.00	0.55 ± 0.07	0.92 ± 0.02	0.11 ± 0.03
	RH2	–	0.33 ± 0.12	0.42 ± 0.01	0.90 ± 0.04	0.18 ± 0.01	0.73 ± 0.09	0.21 ± 0.03
	RH3	–	0.26 ± 0.05	0.55 ± 0.23	0.89 ± 0.01	0.21 ± 0.02	0.73 ± 0.10	0.23 ± 0.08
	IH1	–	0.63 ± 0.11	1.44 ± 0.32	0.92 ± 0.01	0.40 ± 0.09	0.77 ± 0.10	0.18 ± 0.04
	VH1	–	0.67 ± 0.09	1.50 ± 0.43	0.93 ± 0.00	0.40 ± 0.05	0.60 ± 0.20	0.28 ± 0.11
Terbutylazine	RH1	–	0.45 ± 0.06	2.25 ± 0.11	0.92 ± 0.00	0.59 ± 0.12	0.85 ± 0.14	0.12 ± 0.04
	RH2	–	0.17 ± 0.02	1.17 ± 0.32	0.91 ± 0.01	0.19 ± 0.04	0.77 ± 0.09	0.18 ± 0.04
	RH3	–	0.34 ± 0.04	0.46 ± 0.15	0.81 ± 0.08	0.29 ± 0.09	0.58 ± 0.09	0.29 ± 0.01
	IH1	–	0.61 ± 0.08	2.16 ± 0.68	0.91 ± 0.01	0.45 ± 0.13	0.93 ± 0.04	0.09 ± 0.02
	VH1	–	0.50 ± 0.04	1.48 ± 0.36	0.89 ± 0.01	0.41 ± 0.02	0.48 ± 0.34	0.42 ± 0.27

cropping systems. However, there were two exceptions. For metazachlor, a significantly higher K_f was obtained for RH1 columns than for IH1. In addition, for flufenacet, a higher K_f was determined for IH1 than for VH1. Between pesticides, similar trends were observed for the three cropping systems. The highest K_f values were reached for terbuthylazine with 1.48–2.25 cm³/g, chlorotoluron with 1.57–1.88 cm³/g and S-metolachlor with 1.44–1.71 cm³/g. In contrast, the pesticides with the lowest K_f were bentazone with 0.17–0.22, metazachlor with 0.29–0.54 and metamiton with 0.83–1.15 cm³/g. For RH2, terbuthylazine and flufenacet have the highest K_f of 1.17 and 1.07 g/cm³. For RH3, the K_f of flufenacet and S-metolachlor were higher than the other pesticides with 0.77 and 0.55 cm³/g, respectively. Bentazone remained last for RH2 and RH3 with 0.10 and 0.15 cm³/g respectively.

Regarding the DT₅₀ half-life of the pesticides, the DT₅₀ of RH1 is higher than that of RH2 and/or RH3, except for metazachlor, S-metolachlor and ethofumesate. No significant differences were observed between the cropping systems for any of the pesticides. Bentazone is the only pesticide that differs significantly from the others for all columns. Its half-life was significantly higher than the other pesticides, ranging from 18.2 to 48.2 days for the three cropping systems in horizon 1 (RH1, IH1 and VH1), 68.6 days for RH2 and 20.8 days for RH3. However, similar trends were found between the columns. The highest DT₅₀, except for bentazone, were obtained for metamiton and terbuthylazine in all three cropping systems with 1.6–2.1 and 1.1–1.6 days respectively. Flufenacet had the lowest DT₅₀ with 0.5–0.8 days. For RH2 and RH3, metazachlor had the second highest DT₅₀ (6.2 and 10.6 days). Ethofumesate and flufenacet had the lowest DT₅₀ with 1.4–1.5 and 2.0–2.2 days respectively.

Finally, the mass transfer coefficient ω_s shows no difference between cropping systems for the majority of the pesticides. For metazachlor, S-metolachlor and ethofumesate, a higher ω_s was obtained for columns RH1 than IH1 and/or VH1. The ω_s between horizons was similar for bentazone, metamiton and flufenacet. For the other pesticides, a higher ω_s was obtained in horizon 1 than in the other two.

4. Discussion

4.1. Influence of depths on pesticide leaching

The results revealed that the differences in soil properties and structure between the three horizons resulted in more contrasting leaching behaviour than between the production systems.

Horizon 1 (0–30 cm): Despite a larger amount of outflow water and a higher K_s , greater pesticide retention is observed at the surface. The K_f obtained is systematically higher than for the deeper horizons, as reported in the literature (Deng et al., 2017; Felding, 1997). Pesticide adsorption by organic matter (OM) is the main mechanism of pesticide retention in soils when OM is greater than 2 % (Arienzo et al., 1994; Arias-Estévez et al., 2008). The soil in the reference system has 2.5 % OM in horizon 1, compared to only 1 % and 0.7 % for horizons 2 and 3, which explains the high retention of pesticides. Furthermore, horizon 1 is ploughed every year. The cultivation operations create a different structure, changing the properties of the topsoil and the macropores. Consequently, in horizons 2 and 3, a higher percentage of pesticides is recovered from the columns. However, at the end of the experience only three pesticides were significantly more adsorbed at the surface. This may be due to the higher percentage of degradation at the surface for the majority of pesticides with lower DT₅₀. In fact, microbial activity is enhanced in the topsoil, at the root-soil interface, where pesticides are retained for longer, which could explain their rapid degradation (Mills et al., 2001; Rodríguez-Cruz et al., 2006).

Horizon 2 (30–60 cm): The peak of bentazone is reached earlier in horizon 2 than in horizon 1 at similar concentrations. In addition, the percentage of mobile zone (macropores) and the dispersivity are greater in horizon 2 than in the other two horizons, reflecting a greater degree of preferential flow (Koestel et al., 2013). These results can be attributed to

the large number of earthworm galleries and root holes encountered during the field investigations. Horizon 2 contains a ploughed sole over a few centimetres and is undisturbed, resulting in a greater number of macropores and pore connectivity. Undisturbed soil may increase vertical water and pesticide transfer (Cueff et al., 2020; Soto-Gómez et al., 2019). However, the leached proportions are similar to those in horizon 3 due to greater degradation.

Horizon 3 (60–90 cm): For most pesticides, greater adsorption is observed in horizon 3 than in horizon 2. This may be due to the small amount of water flowing through the soil and the pesticides retained in the immobile zone (Imig et al., 2023b). Horizon 3 is characterised by a lower porosity and transfer coefficient between the two regions, as observed by Pot et al. (2010). In addition, horizon 3 has a higher clay content of 19.2 % compared to 13.8 and 13.1 % for RH1 and RH2 respectively. In fact, Imig et al. (2023a, 2023b) have shown that finer textured soils have greater water retention. Smaller clay particles have a greater specific surface area and reduce herbicide leaching by increasing their adsorption and enhancing their degradation in the soil (Kumari et al., 2020; Mendes et al., 2016; Novak et al., 2001). However, only the K_f of three herbicides increased from RH2 to RH3, demonstrating the complexity of their behaviour in soils.

Thus, the different soil properties and structures between the three horizons have resulted in contrasting pesticide fates. In general, only topsoil transport parameters are studied and presented in databases (Dusek et al., 2015; EFSA, 2015; Lewis et al., 2016). Furthermore, these parameters are usually considered independent of soil depth in modelling to limit the number of values to be estimated (Dusek et al., 2011; Mertens et al., 2009). However, this study shows that transport processes within the lower horizons are not negligible. Applying the values of sorption and degradation parameters from the soil surface to the entire soil profile will lead to an overestimation of pesticide degradation and adsorption. As a result, these predictions will be overly optimistic and will underestimate water contamination. The inclusion of soil properties and different soil depths in risk studies would allow more appropriate recommendations to be made on pesticide selection, dose adjustment and application periods.

However, these results must be interpreted with appropriate caution. They are based on three replicates, and the limited size of the soil columns may also depend in part on the spatial variability of the soil, given its heterogeneous nature. As a result, for some of the modalities, the standard deviations are large. Moreover, this study was carried out under controlled laboratory conditions and does not entirely reflect the complexity of natural systems. Investigating these parameters for different horizons under field conditions with more samples and considering them in pesticide leaching risk assessment and decision making is therefore essential. In addition, it should be noted that the soil profile represents a continuum between the three horizons and the quantities of pesticides entering horizons 2 and 3 will depend on the adsorption and degradation of upper horizons.

4.2. Influence of cropping systems on pesticide leaching

Agricultural practices such as tillage are known to generate different soil properties, which can influence the behaviour of pesticides in the soil (Katagi, 2013; Kodešová et al., 2011). The results indicate that production systems can generate different water fluxes and pesticide leaching dynamics.

ICLS system: In the ICLS system, the higher bentazone peak indicates a faster elution of the pesticide. The proportion of the mobile zone is higher, reaching 10 % for the largest average BD, followed by the vegan and reference systems. Thus, despite a denser soil, the larger percentage of macropores generates a greater degree of preferential flow. These preferential flows may be generated by the large fibrous roots of maize as observed by Jin et al. (2018). Indeed, different root architectures and growth stages can strongly influence the hydraulic properties of soils, especially by clogging pores, breaking aggregates or

forming macropores (Ajayi et al., 2019; Lu et al., 2020). The greater connectivity of the pores in the ICLS system results in faster vertical flow, leading to a higher risk of groundwater contamination (Soto-Gómez et al., 2019).

Vegan system: Half of the pesticides eluted faster in the vegan system compared to the other systems. Nevertheless, the concentrations observed in the effluents are lower. The differences in C_{peak} between the systems were generally not significant due to the large heterogeneity between the three replicates, as observed between different columns by Varvaris et al. (2021a, 2021b). Very low water infiltration was observed for replicates 2 and 3. The resulting K_s has a standard deviation twice as high as the other systems. This can be explained by the higher clay content of 17.2 % in the vegan system compared to 13.8 and 12.3 % in the reference and ICLS systems, resulting in lower leaching and a larger specific adsorption surface (Imig et al., 2023a). This high water retention for the vegan system had already been observed by Pirlot et al. (2024) in a study of water retention dynamics on the same plots. With low water flows, pesticides can also be retained longer in the immobile zone. In addition, unlike maize and beet, camelina was harvested and the stubble ploughed one week before the columns were collected. These agricultural practices may have disturbed soil structure and surface macropores, reducing pore hydraulic connectivity and soil hydraulic conductivity through the creation of a more impermeable layer (Alletto et al., 2015; Cueff et al., 2020). Post-harvest rainfall may also have destroyed some low-stability aggregates, leading to sealing of the soil surface (Holland, 2004) as observed for replicates 2 and 3 where a water layer was constantly visible on the surface. Finally, camelina has a fine root system, with 82 % found in the first 30 cm (Berti et al., 2016; Gesch and Johnson, 2015). Fine roots can block soil pores, leading to reduced infiltration capacity (Lu et al., 2020). These low water fluxes resulted in longer pesticide/soil contact times and higher adsorption percentages for most pesticides. This higher adsorption may also be due to the higher clay content and CEC (Porfiri et al., 2015). This pesticide retention resulted in low effluent concentrations and truncated peaks, explaining the apparent early T_{min} and T_{peak} (Cueff et al., 2020). Dispersivity is also higher for VH1 and is almost double that of RH1 and IH1, showing that herbicide transport processes are highly heterogeneous (Vincent et al., 2007). Pesticide transfer is therefore strongly limited in the vegan system due to different soil properties and hydraulic discontinuity caused by agricultural operations.

Reference system: A greater hydraulic conductivity was observed with a higher K_s and mobile/immobile transfer coefficient than the other systems. Although there was higher porosity, the percentage of mobile zone (macropores) was lower, resulting in a reduced degree of preferential flow in comparison to the ICLS system. However, the system showed greater pore connectivity than the vegan system. Sugar beet has a main pivot root located below the plant. As a result, there are few inter-plant roots, resulting in a different soil structure compared to that of maize or camelina. Jin et al. (2018) showed that water infiltration behaviour in soils is significantly affected by root morphological variation between crops.

Differences between systems were mainly related to crops planted, surface tillage and soil properties. However, for most of the pesticides, no significant difference was observed between the percentages of pesticides recovered and degraded or between the K_f and DT_{50} values. The production systems were only implemented for one year and the tillage practices were similar. As a result, they are probably not yet sufficiently differentiated to produce more contrasting leaching behaviour. However, the variations observed show that the production systems could have an impact in the longer term and be a lever for reducing pesticide leaching and groundwater pollution. Future studies should incorporate long-term monitoring of pesticide behaviour under different innovative production systems to assess the cumulative effects and seasonal variations in leaching dynamics. These studies could provide the basis for policies to encourage the adoption of more sustainable practices. These results must be interpreted with appropriate caution, given the small

number of replicates and the limited size of the soil columns, which in some cases resulted in a large standard deviation.

4.3. Influence of pesticide properties

The properties of the pesticides themselves played a crucial role in their leaching behaviour within the same soil. In addition, the mass balances obtained do not always agree with the pesticide transport parameter values obtained. Different pesticide classifications are obtained depending on cropping systems and soil depth, highlighting the complex interaction between pesticide properties and soil characteristics.

Bentazone is the most rapidly eluted herbicide and appears tracer-like with water flow, as observed by De Wilde et al. (2009). With the low value of the K_f , its BTCs indicate a high degree of preferential flow. Koestel et al. (2012) showed that moderate to high preferential flow is found with a clay content above 8–9 %. Thus, the clay content of the soils studied, ranged from 12.3 to 19.2 %, results in a structured soil with stable aggregates and large pores (Bromly et al., 2007; Porfiri et al., 2015). Moreover, preferential flows may be promoted during heavy rainfall applied to the columns (ponding conditions), leading to near saturation conditions (Koestel et al., 2013; Pot et al., 2005). Between replicates, the spatial heterogeneity in the distribution of voids and macropores results in similar values, except for the peak. In fact, agricultural practices and biological activity generate heterogeneous degrees of preferential flow (Schwen et al., 2011; Singh et al., 2018). Bentazone shows very low adsorption (less than 1 %) and high elution (more than 70 %). These results are close to those of Boivin et al. (2004) for a silty soil after 60 days. Its K_f is the lowest and its field DT_{50} the highest, showing its high potential for leaching and groundwater contamination. These results can be explained by its high solubility in water and its weak acid nature, resulting in very low adsorption at pH close to neutral (Arias-Estévez et al., 2008; Boivin et al., 2004). The high quantities of bentazone applied to soils and leached into water can have negative impacts on non-target organisms and human health, with moderate toxicity for birds and humans (Lewis et al., 2016).

Metazachlor and metamitron are the two fastest eluted herbicides after bentazone. As in the PPDB (Lewis et al., 2016), their K_f are ranked just after bentazone. However, despite similar leaching patterns, their mass balances and soil fates are very contrasted. The percentages of metazachlor adsorbed are the lowest after bentazone in all columns, in contrast to metamitron, which is intermediate. In addition, the highest percentages of pesticide recovery in leachates were those of metamitron for topsoil (15.5–36.3 %) and metazachlor (59.8–64.3 %) at depth. In topsoil, the amounts of metamitron degraded are the lowest after bentazone, resulting in high leaching, in contrast to metazachlor, which shows the highest degradation percentages after flufenacet. These results could be attributed to the solubility of metamitron, which is almost four times higher than that of metazachlor (see Table 2). They are in agreement with those of Mamy and Barriuso (2005) who found a low adsorption of these two pesticides in soils, with a higher adsorption and a longer half-life for metamitron than for metazachlor at the surface. Thus, metazachlor appears to be more susceptible to topsoil degradation than the other pesticides. At depth, however, the opposite trend was observed, with very little degradation of metazachlor, which ranked just behind bentazone, resulting in high leaching. Like bentazone, metamitron showed no difference in degradation or leaching between the different depth or cropping system. The mobility of metamitron in soils therefore seems to be less dependent on soil properties and biological activity than other herbicides, as observed by Cox et al. (1996). In addition, the adsorption of metazachlor appears to be more dependent on clay content than on OM content in the soil, with greater adsorption on RH3 and VH1. Both pesticides can cause moderate ecotoxicological risks to birds, fish, daphnia, bees and earthworms. Metazachlor is potentially carcinogenic for humans and may affect reproduction and development. Metamitron presents a high risk to humans as an

endocrine disruptor (Lewis et al., 2016). It is therefore essential to determine the fate of these pesticides in soil and groundwater.

Terbuthylazine and flufenacet are the most strongly retained herbicides in soils. According to the PPDB, their K_f are the two highest. However, in our study these two pesticides show very different behaviors. The K_f of terbuthylazine is the highest for RH1, RH2 and IH1, but only the third highest for RH3 and VH1. The percentages adsorbed are among the three largest, but only first for RH1. Its adsorption therefore seems to depend more on the OC than on the clay content, as observed by (Kodešová et al., 2011). Terbuthylazine is a weak basic herbicide with very low solubility, explaining its low leaching. Terbuthylazine is highly toxic to humans and birds. Flufenacet is also highly toxic to humans as an endocrine disruptor.

The K_f of flufenacet is more variable, ranging from 1st for RH3 to 5th for VH1. In the topsoil, its adsorption percentages are the third lowest after bentazone and metazachlor. However, at depth it is one of the most adsorbed pesticides. Flufenacet showed greater adsorption in the RH3 and VH1 columns than in the others. Its recovery percentages are the lowest of the 8 pesticides and its degradation percentages are the highest for topsoil and the second lowest and highest at depth, after ethofumesate. Thus, flufenacet, like metazachlor, seems to be more sensitive to degradation than the other pesticides, but for all columns, and probably shows a greater affinity to clays than to OC.

Ethofumesate has the lowest recovery in the leachates at depth (12.6–16.7 %) due to higher degradation (79.5–86.7 %) than the others, and one of the lowest at surface due to high adsorption (12.1–29.7 %). It shows a similar leaching and degradation percentage for all columns as bentazone and metamitron. Its adsorption seems to depend only on the OC content, as it shows a very low adsorption (0.73 %) for RH3 compared to 29.7 % for VH1, as shown by Facenda et al. (2024). Its low leaching may be due to its low water solubility, which is close to that of flufenacet. Chlorotoluron has the highest adsorption percentages for most columns (7.43 to 31.6 %), with the highest values found for IH1 and VH1. The K_f values are among the three highest (1.57–1.88 cm³/g for horizon 1 and 0.50 cm³/g for horizon 2), except for RH3. These results do not agree with the PPDB ranking, where k_f is among the lowest, just after metamitron. Kodešová et al. (2011) found large adsorption related to OM content with higher K_f for terbuthylazine, followed by chlorotoluron and finally s-metolachlor. However, the K_f rankings obtained in our study vary greatly depending on the soil depth and cropping systems.

Thus, the properties of pesticides have a major influence on their fate within the same soil. More importantly, the leaching or adsorption rates are not always in agreement with the K_f values obtained or with those of the PPDB. Different pesticide classifications are obtained depending on cropping systems and soil depth. These results demonstrate the

importance of studying the fate of all pesticides of environmental concern in the soils of interest to manage them more effectively. These pesticides can be highly toxic to non-target organisms such as birds and earthworms, as well as to human health. Pesticides remaining in soils can have an impact on microbial growth and enzymatic activity, which can affect their biodegradation (Wojcik et al., 2020). In large concentrations, some pesticides can be toxic and reduce soil microbial biomass, affecting respiration, degradation of organic matter and nutrient cycling (Rose et al., 2016). On the other hand, their presence can also favour microorganisms capable of degrading them, to the detriment of other microorganisms, disrupting the established balance (Wojcik et al., 2020). Finally, the application of herbicides can lead to lower plant diversity and exudate production, and can therefore reduce the diversity and activity of soil organisms. Identifying the fate of pesticides in groundwater and soil is therefore crucial for the future.

4.4. Comparison with database

Tables 10 and 11 compare the transport parameters K_f and β , and field DT₅₀ from the PPDB, the EFSA peer review for the Walloon soil-climate zone and the values used in Belgium with the results for the surface reference system.

In comparison with the databases like the PPDB or EFSA, the results indicate important differences in transport parameters for most pesticides, which may lead to underestimation of pesticide and metabolite leaching.

The K_f values obtained are within the PPDB and EFSA ranges. These ranges are indeed very broad as they are based on different studies and soils from all over Europe. Typical laboratory PPDB values are similar to our results for metazachlor and ethofumesate. They are lower for metamitron, S-metolachlor and chlorotoluron, with a larger difference for s-metolachlor. The PPDB value for s-metolachlor is the lowest of the eight herbicides studied, while our results show a high K_f , ranking third. Conversely, flufenacet and bentazone have higher PPDB K_f values. However, the EFSA values used in Belgium are all much higher than those obtained in this study. The largest differences were found for bentazone, S-metolachlor and flufenacet. The assumption of higher adsorption may lead to an underestimation of pesticide leaching and therefore groundwater contamination. Indeed, for most of the pesticides assessed in this work, EFSA concluded that the potential for groundwater contamination was low based on the 9 FOCUS scenarios, in particular for bentazone, metazachlor, chlorotoluron and terbuthylazine. Groundwater analyses in Belgium identify bentazone, chlorotoluron and terbuthylazine as major contaminants (SPW ARNE, 2022). This tends to confirm our results. These large differences can be due to different methodologies for determining transport parameters. The

Table 10

Comparison of the K_f values and associated β values in brackets of the PPDB, the EFSA peer review for the Walloon soil-climate zone and the values used in Belgium with the results of this study for the surface reference system. For individual values, uncoloured boxes represent values similar to those obtained, less than 1.5 times higher, yellow boxes correspond to 1.5 to 2 times higher, orange boxes 2 to 3 times higher and red boxes more than 3 times higher. Ranges are shown in red if they do not include the value reported in this study. The blue boxes indicate values below the results.

	K_f (ml/g) (β)	Bentazone	Metamitron	Metazachlor	Metolachlor	Ethofumesate	Chlorotoluron	Flufenacet	Terbuthylazine
Results	RH1	0.22 (1.00)	1.15 (0.75)	0.94 (0.92)	1.71 (0.92)	1.59 (0.90)	1.88 (0.92)	1.66 (0.89)	2.25 (0.92)
PPDB	Typical lab	0.97 (0.93)	1.09 (0.81)	1.02 (0.99)	0.93 (0.89)	1.74 (0.91)	1.30 (0.88)	4.38 (0.92)	5.10 (0.93)
	Range	0.02-3.06 (0.70-1.03)	0.36-5.88 (0.67-0.95)	0.37-2.20 (0.99-1.00)	0.07-2.57 (0.55-1.35)	0.70-6.20 (0.84-0.96)	0.78-2.74 (0.83-0.93)	1.48-10.6 (0.88-0.98)	2.10-10.5 (0.88-0.98)
EFSA	Range	0.08-1.94 (0.70-1.00)	0.56-4.65 (0.64-0.85)	0.48-3.50 (0.82-0.85)	0.59-4.80 (0.91-0.93)	0.70-6.20 (0.82-0.96)	2.70-9.60 (0.83-0.93)	1.00-14.7 (0.84-0.98)	2.10-10.5 (0.88-0.98)
Belgium	Mean value	0.76 (0.97)	3.06 (0.77)	2.75 (0.84)	5.33 (0.93)	2.95 (0.90)	4.52 (1.00)	5.05 (0.87)	5.78 (0.90)

Table 11

Comparison of the DT₅₀ values of the PPDB, the EFSA peer review for the Walloon soil-climate zone and the values used in Belgium with the results of this study for the surface reference system. For individual values, uncoloured boxes represent values similar to those obtained, less than 2 times higher, yellow boxes correspond to 2 to 10 times higher, orange boxes 10 to 25 times higher and red boxes more than 25 times higher. Ranges are shown in red if they do not include the value reported in this study. The blue boxes indicate values below the results.

	DT ₅₀ (days)	Bentazone	Metamitron	Metazachlor	Metolachlor	Ethofumesate	Chlorotoluron	Flufenacet	Terbuthylazine
Results	RH1	18.2	2.1	1.3	1.4	1.5	1.4	0.8	1.6
PPDB	Typical lab	20.0	19.0	10.8	15.0	21.6	33.5	19.7	72.0
	Range	8.0-35.0	2.2-44.5	5.5-25.3	8.0-38.0	8.5-80.5	10.3-144	7.04-37.4	38.2-167
EFSA	Range	3.0-6.9	6.6-21.7	5.1-14.4	3.6-55.7	13.1-110	26.0-42.0	13.1-54.2	6.9-35.7
Belgium	Mean value	4.7	11.1	8.5	22.4	40.8	42.0	27.7	19.5

parameters in the databases are generally determined by batch adsorption in flasks, at equilibrium. These experiments are generally of short duration and maximise contact between the soil and the pesticides through agitation. Therefore, they do not take into account soil structure or preferential flow. Furthermore, as will be seen in this article, soil type and soil properties have a major influence on the fate of pesticides in soil, which can lead to significant differences in the estimated parameters. Cueff et al. (2020) also found significant differences between the transport parameters estimated for several pesticides and the PPDB. They showed that the K_d values obtained for the same soil in batch experiments did not correspond to the retardation factor obtained in undisturbed soil columns due to the soil structure.

The DT₅₀ obtained for bentazone is in good agreement with the typical PPDB value. However, it is much higher than the value used in Belgium and the EFSA range. This indicates an overestimation of its degradation and therefore a possible underestimation of its leaching and contamination of groundwater. An overestimation of adsorption and degradation could have important environmental consequences for bentazone, which is very poorly retained in soils. For the other herbicides, very low DT₅₀ values were obtained. They are lower than the ranges, PPDB values and values used in Belgium. The use of higher DT₅₀ values leads to a possible overestimation of the leaching of these pesticides, which has less impact on the environment than the opposite for decision making purposes. However, this may lead to underestimating metabolite formation. The largest differences were found for ethofumesate, chlorotoluron and flufenacet, followed by terbuthylazine and s-metolachlor. Metabolites are generally more mobile and persistent than parent herbicides, causing significant leaching and water contamination (Kiefer et al., 2019; Lewis et al., 2016). According to EFSA studies, the metabolites of terbuthylazine, s-metolachlor and metazachlor have a high leaching potential (Álvarez et al., 2023; Brancato et al., 2017; Lentdecker et al., 2017). Along with these, metabolites of metamitron and flufenacet were classified as highly mobile in the PPDB, with high DT₅₀ of up to 400 days for metolachlor ESA (Lewis et al., 2016). An investigation of novel metabolites detected in groundwater showed that those of terbuthylazine, metamitron, metazachlor and flufenacet were found at higher concentrations than the parent pesticides (Kiefer et al., 2019). It is therefore important not to underestimate the degradation of pesticides in soils. Special emphasis should be placed on the study of the fate of metabolites in soils and their measurement during monitoring, which is rarely investigated and all too often neglected.

Several sensitivity analyses have already been carried out on solute fate using Hydrus 1-D. Jian et al. (2024) showed that the adsorption coefficient K_d and the dispersivity λ are the two main parameters influencing the migration of naphthalene in soils. The concentration of the peak is the most sensitive to the concentration of the influent, followed by K_d, first order degradation rate coefficient (μ) and λ. Cheviron

and Coquet (2009) also carried out a sensitivity analysis on the fate of a pesticide using the dual porosity mobile-immobile model. They found a very high sensitivity of pesticide sorption and degradation parameters, starting with Freundlich exponent β, then K_f and finally μ. Differences in the use of adsorption and degradation parameters can therefore have a major impact on the model outputs and hence on the assessment of the potential risks of water contamination. In addition, the limitation of the dual-porosity model is that it assumes that the flow of water is limited only to the mobile region (macropores). Thus, water does not move at all in the immobile zone or soil matrix, which can only exchange, retain and store water without convective flow (Šimůnek et al., 2003). In reality, water flows in both regions of the soil, but at different rates. Some researchers even consider triple soil porosity to improve the modelling (Varvaris et al., 2021a). This limitation can lead to different behaviour of water and solute flows between reality, the database and the model values.

These results highlight the need to adapt the K_f and DT₅₀ values for all pesticides and metabolites of environmental concern to the soil type studied for future modelling and decision-making. Parameter values from databases such as the PPDB or EFSA studies should be used with caution and are not always reliable. These results are in agreement with the recent study by Mamy et al. (2024) who modelled the leaching of many pesticides in soils under more sustainable agricultural practices, using three different models. They used generic transport parameters from the PPDB as used for regulatory risk assessment. They found that the models performed poorly in reproducing the concentration of pesticides in the leachates, with underestimation in most cases.

In addition, unmonitored metabolites pose a risk to water quality. Plant protection products should therefore be used more cautiously, given the difficulty and uncertainty of assessing their fate in soil and water. Studies should include the behaviour of metabolites in soils to provide a more complete picture of the potential for pesticide contamination. These products also represent a threat to microorganisms, animals and human health, particularly for farmers with high levels of exposure.

5. Conclusion

In this study, the fate of eight pesticides of concern to groundwater was investigated under three production systems and at three soil depths. Leaching experiments in undisturbed columns, mass balances and inverse modelling using Hydrus 1-D to adjust mobility parameters were carried out. The results provide a better understanding of how cropping systems and soil properties can influence the behaviour of different pesticides in soils and highlight potential risks to the environment.

The results revealed that the differences in soil properties and

structure between the three horizons resulted in more contrasting leaching behaviour than between the production systems. The soil surface retains more pesticides due to its higher organic matter content and biological activity, which enhances degradation. Horizon 2 is characterised by greater preferential flow due to the presence of more earthworm galleries and root channels, as a result of undisturbed soil. Pesticide adsorption increases in horizon 3 due to higher clay content and lower water flow, resulting in longer retention in the immobile zone. These results highlight the need to consider deeper soil horizons in pesticide leaching risk assessment, as transport processes and retention characteristics vary significantly with depth. Applying transport parameter values from the soil surface to the entire profile will result in overly optimistic predictions and will lead to an underestimation of water contamination.

Our analysis demonstrates that cropping systems can affect pesticide leaching dynamics. The findings highlight the importance of root architecture, soil properties and surface tillage in influencing water flow and pesticide transport through soil. However, the short implementation period of the production systems probably resulted in insufficient differentiation and prevented significant differences in recovery rates, degradation rates, K_f or DT_{50} values. Thus, in the long term, different agricultural systems could have a significant effect on pesticide leaching and be a lever for reducing groundwater pollution.

The properties of the pesticides themselves played a crucial role in their leaching behaviour within the same soil. In addition, the mass balances obtained do not always agree with the pesticide transport parameter values obtained or with those of the PPDB. Different pesticide classifications are obtained depending on cropping systems and soil depth, highlighting the complex interplay between pesticide properties and soil characteristics. These results demonstrate the importance of studying the fate of all pesticides of environmental concern in the soils of interest to manage them more effectively.

When compared with established databases such as the PPDB and the EFSA peer review for Europe, our results indicated discrepancies in transport parameters for most pesticides. The EFSA values used in Belgium show higher K_f than our results, which could lead to an underestimation of pesticide leaching and water contamination. In addition, except for bentazone, higher DT_{50} values were found in the databases, which could lead to an underestimation of the production and leaching of metabolites, which are more mobile and persistent than the parent pesticides. Therefore, parameters derived from databases should be used with caution and are not always reliable. Hence, this study shows the need to investigate the fate and adapt the transport values of all environmentally problematic pesticides to site-specific characteristics.

Future studies should incorporate long-term field monitoring of pesticide behaviour under different innovative production systems to assess the cumulative effects and seasonal variations in leaching dynamics. This could provide a more comprehensive understanding of pesticide persistence and mobility over time. The impact of sustainable agricultural practices should also be studied for other types of soil, with varying characteristics. Research should include a wider range of pesticides and their metabolites to help understand the leaching behaviour of different classes of chemicals and their environmental impact. In addition, a special emphasis should be placed on studying the fate of metabolites in soils. Developing advanced risk assessment models that account for soil depth variability and site-specific conditions is critical to improve predictions of groundwater contamination and inform better decision-making. This research provides some insight for developing more sustainable agricultural practices that can minimize environmental contamination while ensuring effective pest management. Such knowledge is essential for advancing policies and practices aimed at protecting environmental safety and promoting the resilience of agricultural ecosystems in the face of ongoing environmental and regulatory challenges.

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CRedit authorship contribution statement

Clémence Pirlot: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alodie Blondel:** Writing – review & editing, Resources, Methodology, Investigation. **Boris Krings:** Resources, Methodology, Investigation. **Bastien Durenne:** Writing – review & editing, Resources. **Olivier Pigeon:** Supervision, Resources. **Aurore Degré:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available on request from the corresponding author.

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